

GAN-Based Document Restoration with Frozen HTR Recognizer and Multi-Component Loss Optimization for Historical Manuscripts

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Abstract—Historical handwritten documents suffer from various degradation artifacts that significantly impair Handwritten Text Recognition (HTR) systems. While conventional document restoration methods focus on visual quality, they often fail to preserve text readability for HTR tasks. This paper proposes an HTR-oriented document restoration framework based on Generative Adversarial Networks (GAN) with two key contributions: (1) integration of a frozen pre-trained HTR recognizer that provides stable text-aware gradients without joint training instabilities, and (2) systematic multi-component loss function optimization through ablation studies. Unlike conventional approaches that rely solely on image-level supervision, our method explicitly optimizes text readability through gradients from the frozen recognizer. We conduct quantitative evaluation on a realistic semi-synthetic dataset constructed from authentic 16th-18th century paleographic documents from the National Archives of the Republic of Indonesia (ANRI) with controlled degradation augmentation to enable paired ground-truth evaluation. On 712 test images, our framework achieves a Character Error Rate (CER) of 34.9%, reducing CER from 83.4% (degraded condition) with 58.2% relative improvement, while maintaining high visual quality (PSNR 30.74 dB, SSIM 0.987). This performance approaches the theoretical maximum on clean images (CER 34.1%). Qualitative validation on 15 real ANRI manuscripts with authentic historical degradation demonstrates effective generalization to damage patterns that cannot be replicated synthetically. Ablation studies reveal that a 4-component loss configuration (pixel, adversarial, perceptual, CTC) achieves optimal balance, while dual-modal discriminator exploration shows marginal contribution (Δ PSNR +0.28 dB, $p > 0.05$), confirming that recognizer freezing strategy and loss optimization are more critical than discriminator architectural complexity.

Index Terms—Generative Adversarial Networks, Document Restoration, Handwritten Text Recognition, Loss Function Optimization, Historical Document Processing, Deep Learning.

I. INTRODUCTION

HISTORICAL handwritten documents constitute invaluable cultural heritage, containing critical information about past events, governance, trade, and social structures. The National Archives of the Republic of Indonesia (ANRI) preserves an extensive collection of colonial-era documents

from the 16th to 18th centuries, including materials recognized by UNESCO as part of the Memory of the World Register.

These documents, written in Dutch paleographic script during the VOC (Vereenigde Oostindische Compagnie) and Dutch East Indies periods, provide unique insights into Indonesia's historical development. However, these manuscripts have suffered severe degradation including (Figure 1): ink bleed-through, iron gall ink corrosion, paper aging, physical damage, and fading ink.

Such degradation significantly impairs text readability and poses substantial challenges for Handwritten Text Recognition (HTR) systems.

These degradation artifacts severely disrupt HTR system performance, which is essential for large-scale digitization, transcription, search, indexing, and information extraction from historical archives. Despite substantial advances in HTR technology using deep learning approaches, particularly Convolutional Recurrent Neural Networks (CRNN) and transformer-based architectures, these systems remain highly sensitive to image quality and degradation levels.

A. Limitations of *yang ada* Approaches

Degraded historical documents present unique challenges for HTR systems, as noise, stains, blur, and background degradation can significantly reduce recognition accuracy. Automated document restoration aims to enhance image quality before HTR processing, yet *yang ada* approaches have fundamental limitations:

First, conventional document restoration methods (including classical thresholding such as Otsu [1] and Sauvola [2], as well as deep learning approaches based on U-Net and GAN) optimize purely visual metrics (PSNR, SSIM) without considering the impact on HTR readability. Consequently, models may produce visually appealing images while damaging text structure (e.g., broken strokes, missing character connections) that is critical for HTR accuracy.

Second, HTR integration in document restoration has been proposed by Souibgui *et al.* [8] through joint training of generator and recognizer. While innovative, this approach uses a single-modal discriminator that only evaluates visual quality, and joint training can cause optimization instabilities as the generator and recognizer compete for conflicting gradients.

Third, an evaluation gap exists—most benchmarks (DIBCO series) only provide visual ground truth without text transcriptions, making HTR impact validation infeasible without addi-

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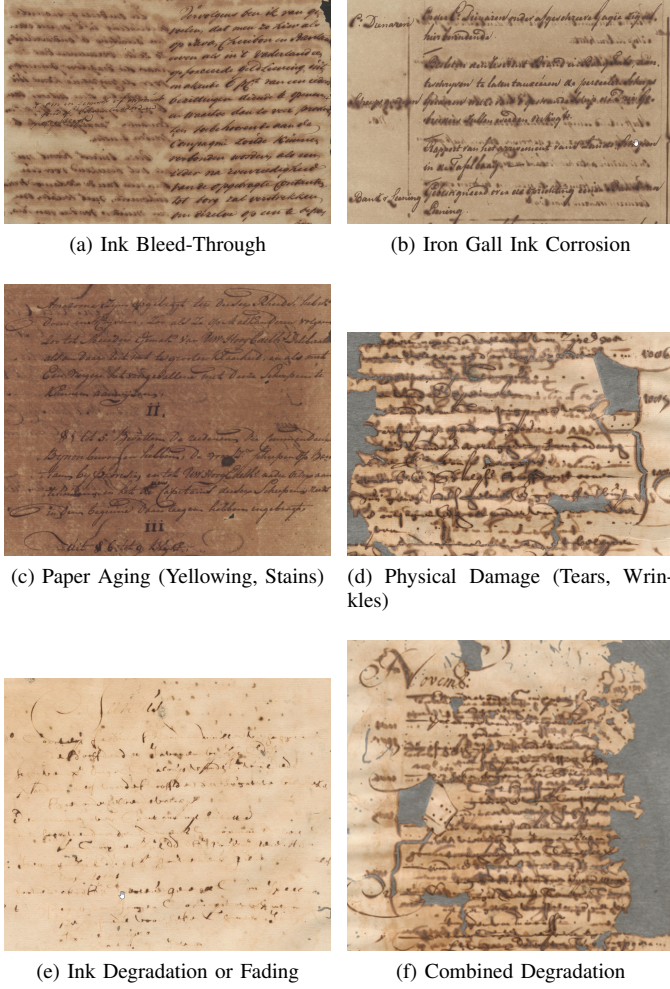


Fig. 1: Examples of degradation in 16th-18th century ANRI documents: (a) ink bleed-through, (b) iron gall ink corrosion, (c) paper aging, (d) physical damage, (e) fading ink, (f) combined degradation. Images demonstrate the varied degradation levels that challenge HTR systems.

tional annotations. Datasets for 16th-18th century paleographic documents with authentic degradation are limited.

This research addresses these limitations through: (1) frozen HTR recognizer integration for stable text-aware guidance, (2) systematic loss function optimization via ablation studies, (3) comprehensive evaluation on authentic ANRI paleographic manuscripts, and (4) empirical validation of architectural design decisions including dual-modal discriminator contribution analysis.

B. Research Contributions

This research proposes an HTR-oriented document restoration framework that explicitly optimizes text readability while maintaining visual quality. Based on empirical validation through systematic ablation studies, the research contributions are organized as follows:

Main Contributions:

- 1) **Frozen HTR Recognizer Strategy:** Using a frozen pre-trained HTR recognizer provides stable text-aware gradients without joint training instabilities. Ablation studies demonstrate significant impact on convergence stability ($p < 0.001$) and preservation of recognition capabilities (see Section V-B).
- 2) **Multi-Component Loss Function Optimization:** Identification of an optimal 4-component configuration (adversarial, pixel reconstruction, perceptual, CTC) through systematic ablation studies. This configuration achieves Pareto balance between visual quality (PSNR 30.74 dB) and HTR readability (CER 34.9%), outperforming the 5-component configuration (see Section V-B).

Methodological Contributions and Exploratory Findings:

- 3) **Systematic Ablation Studies:** Evaluation of each architectural and loss component's contribution, providing practical guidance for component selection. Finding: simple configurations with proper alignment achieve better performance than over-complex configurations.
- 4) **Evaluation on Authentic Historical Documents:** Quantitative evaluation on realistic semi-synthetic dataset constructed from authentic 16th-18th century ANRI paleographic handwriting with controlled degradation augmentation ($n=712$), enabling precise measurement with paired ground truth. Qualitative validation on real ANRI manuscripts with authentic historical degradation ($n=15$) demonstrates effective generalization to damage patterns that cannot be replicated computationally.
- 5) **Findings on Dual-Modal Discriminator:** Exploration of discriminator architecture with CNN (visual) and LSTM (textual) pathways reveals marginal contribution to final metrics ($\Delta\text{PSNR} = +0.28$ dB, $\Delta\text{CER} = -0.01\%$, $p > 0.05$, statistically insignificant). This finding confirms that the key factors are training protocol and frozen recognizer strategy, not discriminator architectural complexity (see analysis in Section VI-A).

C. Research Hypothesis and Validation

This research validates the hypothesis that a document restoration framework with frozen HTR recognizer integration and HTR-oriented loss function optimization produces statistically significant Character Error Rate (CER) reduction compared to degraded baseline conditions (H1). Systematic ablation studies measure each component's contribution, with key findings: (1) recognizer freezing strategy proves to be a critical factor for stability and convergence, (2) 4-component loss configuration achieves optimal balance, and (3) dual-modal discriminator shows marginal contribution ($p > 0.05$). These findings confirm that system success primarily depends on recognizer freezing strategy and loss optimization, not discriminator architectural complexity. Complete results and statistical analysis are presented in Section V.

D. Paper Organization

The paper is organized as follows: Section II reviews related work in document enhancement, GANs for image restoration,

and HTR systems. Section III presents the proposed architecture including Enhanced U-Net generator, dual-modal discriminator (CNN+LSTM), frozen recognizer integration, and optimized loss function with adaptive balancing. Section IV describes experimental setup, dataset, training methodology, and hyperparameter optimization strategies. Section V presents comprehensive results including quantitative comparison with degraded baseline input and clean ground truth reference, systematic ablation studies to validate each component's contribution, and qualitative analysis on real historical documents. Section VI discusses findings, theoretical and practical implications, limitations, and future directions. Finally, Section VII concludes the paper with a summary of contributions and research impact.

II. RELATED WORK

This section reviews prior research in three interconnected areas: document image restoration and enhancement, generative adversarial networks for image restoration, and handwritten text recognition systems.

A. Document Image Restoration and Enhancement

Document restoration aims to recover degraded documents by removing noise, artifacts, and background degradation while preserving text readability for OCR/HTR systems or human reading. Unlike binarization which produces binary output (black-and-white), document restoration can generate grayscale or color output that retains pixel intensity information for better visual quality.

1) *Classical Methods: Binarization and Thresholding:* Early approaches to document enhancement focused on binarization using thresholding. Global binarization methods such as Otsu [1] determine a single threshold value for the entire image by maximizing inter-class variance. While computationally efficient, global methods tend to be less effective on documents with varying illumination or complex degradation.

Local adaptive thresholding methods address this limitation by computing pixel-specific thresholds based on local neighborhoods. Sauvola and Pietikäinen [2] proposed a widely-used method that combines local mean and standard deviation. Niblack [3] and Wolf [4] developed similar local approaches with different adaptation strategies. However, these thresholding-based methods produce binary output that loses pixel intensity information and may eliminate fine details such as thin strokes.

2) *Deep Learning Methods for Document Restoration:* Since the mid-2010s, with advances in deep learning and availability of large labeled datasets such as DIBCO [35], document restoration approaches have shifted from binarization to grayscale enhancement that preserves intensity information. U-Net-based auto-encoders [9] and Generative Adversarial Networks (GANs) enable end-to-end learning of mappings from degraded to clean documents with potential to preserve fine details critical for HTR.

B. Generative Adversarial Networks for Image Restoration

GANs [23] have substantially transformed the field of image restoration with their ability to generate realistic outputs through adversarial training between generator and discriminator. Unlike auto-encoders that only minimize reconstruction loss, GANs can produce sharper details and more natural textures.

1) *Conditional GANs for Image-to-Image Translation:* Pix2pix [14] introduced the conditional GAN (cGAN) framework for image-to-image translation tasks, where the generator receives an input image and produces a corresponding output image. The PatchGAN discriminator in Pix2pix evaluates authenticity at local patch level, enabling better capture of high-frequency details. This framework has been successfully applied to various domains including colorization, style transfer, and semantic segmentation.

CycleGAN [15] extends this concept to unpaired image-to-image translation using cycle consistency loss, enabling training without paired data. While useful for domain adaptation, unpaired approaches are less suitable for document restoration where paired degraded-clean data is available and pixel-level accuracy is crucial for preserving text content.

2) *GANs for Document Restoration and Enhancement:* In the context of document restoration, several GAN approaches have been proposed:

DE-GAN [7] uses a U-Net generator with PatchGAN discriminator for document enhancement tasks including deblurring, watermark removal, and background cleaning. While achieving high visual quality, DE-GAN focuses on optimizing pixel-level similarity and does not explicitly integrate HTR readability metrics in its loss function.

DocEnTr [16] introduces a document enhancement transformer that combines self-attention mechanisms with adversarial training. This approach demonstrates capability to capture long-range dependencies, but has high computational cost and has not integrated explicit HTR guidance.

HTR-GAN (Souibgui *et al.*) [8] represents the most relevant work to this research, integrating an HTR recognizer within the GAN framework for document enhancement. The generator and recognizer are jointly trained with combined losses from adversarial, reconstruction, and CTC. The joint training approach potentially causes optimization instabilities as the recognizer and generator compete for gradients—a theoretical concern that motivates the frozen recognizer approach in this research.

Distinctions of Proposed Approach: This research fundamentally differs from HTR-GAN in: (1) Frozen recognizer—we use a frozen pre-trained recognizer for stable gradients, instead of joint training; (2) Dual-modal discriminator—our discriminator processes images (CNN) and text sequences (LSTM) in parallel with potential for visual-textual coherence evaluation (although ablation studies show marginal impact, see Section VI); (3) Adaptive loss balancing—adaptive balancing mechanism that dynamically adjusts ratios between CTC loss and visual losses for optimal balance; (4) Enhanced architecture—residual blocks, attention gates, and spatial attention for better preservation of thin strokes.

C. Handwritten Text Recognition Systems

HTR systems aim to recognize handwritten text from document images. The evolution of HTR has progressed from classical HMM-based methods to modern deep learning architectures.

CRNN (Convolutional Recurrent Neural Network) [17] combines CNN for feature extraction with RNN (typically LSTM/GRU) for sequence modeling, trained with CTC loss for alignment-free recognition. This architecture has become the foundation for many modern HTR systems due to its ability to handle variable-length sequences without explicit segmentation.

Attention mechanisms have improved HTR performance by enabling models to focus on relevant regions when recognizing each character. Encoder-decoder architectures with attention [18] can learn character segmentation and alignment implicitly.

Transformer-based HTR [19], [22] leverages self-attention to capture long-range dependencies without recurrent connections. Vision transformers and hybrid CNN-Transformer architectures demonstrate state-of-the-art performance on various handwriting benchmarks, including historical and modern documents.

Relevance to This Research: HTR systems are sensitive to image quality—degradation such as noise, blur, or broken strokes can increase CER. This research uses a frozen pre-trained hybrid CNN-Transformer recognizer to extract recognition features that guide the generator to produce HTR-readable output. The frozen recognizer approach differs from joint training because: (1) stable gradients—recognizer weights do not change; (2) consistent evaluation—recognizer performance does not degrade during GAN training; (3) modular design—recognizer can be updated independently.

D. Gap Analysis and Positioning

Based on the literature review above, Table I summarizes this research’s position relative to existing approaches in terms of visual quality, HTR awareness, discriminator architecture, and loss optimization. The identified gaps are: (1) exploration of dual-modal (visual + textual) architecture in the discriminator as architectural completeness for bilateral validation—although ablation studies show marginal impact on final metrics (see Section VI), (2) absence of systematic loss weight optimization to balance visual quality and HTR readability, and (3) potential instabilities from joint recognizer-generator training that can be addressed with a frozen recognizer approach, as motivated by theoretical concerns discussed in Section II. Empirical validation of each component’s relative contribution is presented in comprehensive ablation studies (Section V), with Gap #2 (systematic loss weight optimization) showing the most significant impact on final performance.

III. PROPOSED METHOD

This section presents the proposed HTR-oriented document restoration framework. We first provide an architectural overview, then detail each component: the U-Net generator, dual-modal discriminator, frozen HTR recognizer integration, and optimized multi-component loss function.

TABLE I: Comparison of document enhancement approaches based on visual quality, HTR awareness, dual-modal discriminator, and loss optimization with component validation through ablation studies.

Method	Visual Quality	HTR Aware	Dual Disc.	Loss Optim.
Classical [1], [2]	Low	No	No	N/A
U-Net [9]	High	No	No	No
DE-GAN [7]	High	No	No	No
Souibgui <i>et al.</i> [8]	High	Yes	No	No
Proposed	High	Yes	Yes	Yes

A. Framework Overview

To address the limitations of conventional restoration methods that ignore machine text readability, this research proposes an HTR-oriented GAN framework that explicitly integrates a text recognizer in the training loop. The proposed framework, illustrated in Figure 2, operates on degraded handwritten document line images and produces clean, readable output optimized for both visual quality and HTR performance.

The restoration process flow is as follows: A degraded image I_{deg} is input to generator G which produces a restored image I_{gen} . The dual-modal discriminator D receives pairs $(I_{\text{gt}}, \text{text}_{\text{gt}})$ as real samples and $(I_{\text{gen}}, \text{text}_{\text{pred}})$ as fake samples for adversarial learning. In parallel, the frozen HTR recognizer R extracts features from I_{gen} to ensure text readability through CTC loss and recognition feature loss.

The framework consists of four main components:

- 1) **Generator (G):** Enhanced U-Net architecture mapping degraded images I_{deg} to restored images I_{gen} . The generator is equipped with Residual Dense Blocks and Attention Gates for preserving fine text details while removing background noise.
- 2) **Dual-Modal Discriminator (D):** Discriminator with parallel CNN and BiLSTM pathways that assess visual quality and text coherence simultaneously. Bilateral cross-attention fusion mechanism enables visual-textual consistency validation to avoid artifacts that damage character structure.
- 3) **Frozen HTR Recognizer (R):** Pre-trained Transformer-based recognizer with fixed weights providing HTR-aware gradients without catastrophic forgetting risk. The recognizer achieves CER 33.72% on initial ANRI validation dataset ($n=948$), and CER 26.57% (validation set, $n=710$) and 34.1% (test set, $n=712$) on clean ground truth images in this study—differences reflect text complexity variation across dataset splits.
- 4) **Multi-Component Loss Function:** Weighted combination of $\mathcal{L}_{\text{pixel}}$ ($\lambda = 50$), $\mathcal{L}_{\text{adv}}$ ($\lambda = 3$), $\mathcal{L}_{\text{perc}}$ ($\lambda = 1$), $\mathcal{L}_{\text{rec_feat}}$ ($\lambda = 8$), and $\mathcal{L}_{\text{CTC}}$ ($\lambda = 0.15$) optimized through curriculum learning to balance visual quality and text readability.

Unlike HTR-GAN [8] which jointly trains the recognizer with the generator, our method uses a frozen recognizer for training stability. The dual-modal discriminator architecture serves as exploration for bilateral visual-textual validation not

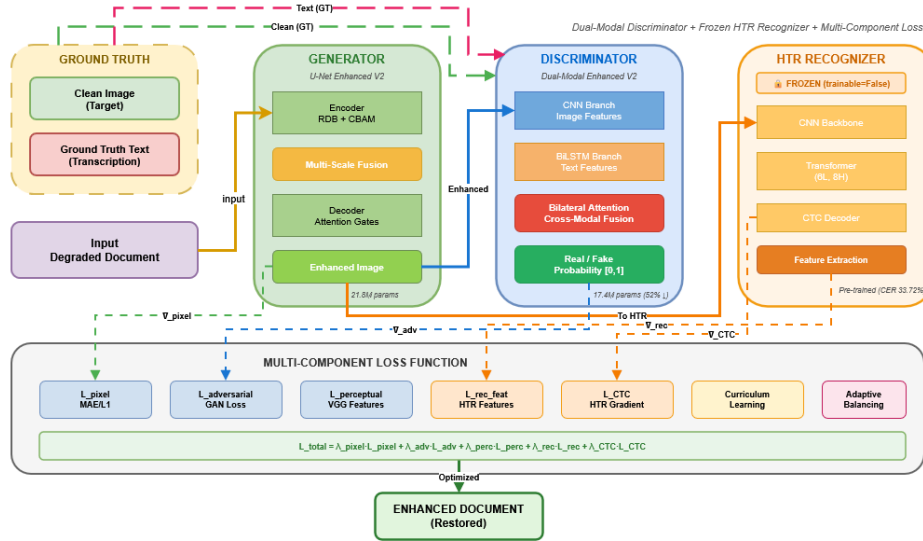


Fig. 2: HTR-oriented document restoration framework with dual-modal discriminator and frozen HTR recognizer. Three main components: (1) Enhanced U-Net Generator for image restoration, (2) Dual-Modal Discriminator (CNN+BiLSTM) for bilateral visual-textual validation, (3) Frozen HTR Recognizer for HTR-aware gradients. Training uses curriculum learning with multi-component loss (pixel, perceptual, adversarial, recognition feature, HTR). Total trainable parameters: 108.3M (Generator 89.4M + Discriminator 18.9M), Frozen recognizer: 27.86M.

present in previous methods that rely only on single visual discriminators.

The training process adopts alternating adversarial strategy with three-phase curriculum learning (detailed in Section III-F):

- Discriminator update: D is trained to distinguish real pairs $(I_{gt}, text_{gt})$ from generated pairs $(I_{gen}, text_{pred})$, with dual-modal supervision on visual and textual domains simultaneously.
- Generator update: G is optimized to fool D while minimizing $\mathcal{L}_{pixel}$ against ground truth, $\mathcal{L}_{perc}$ for perceptual structure preservation, $\mathcal{L}_{rec_feat}$ for HTR feature consistency, and $\mathcal{L}_{CTC}$ for explicit text readability.

B. Generator Architecture: Enhanced U-Net with Residual Blocks

This research adopts an Enhanced U-Net architecture that integrates four techniques for high-quality document restoration. Unlike conventional U-Net that only uses standard convolutional blocks, the Enhanced architecture integrates: (1) Residual Dense Blocks (RDB) [24] for hierarchical feature extraction with better gradient flow, (2) Convolutional Block Attention Module (CBAM) [25] for adaptive feature recalibration on channel and spatial dimensions, (3) Multi-Scale Feature Pyramid (MSFP) [26] with dilated convolutions [27] at the bottleneck for multi-resolution context aggregation, and (4) Attention Gates [28] in the decoder for selective focus on text-bearing regions while suppressing background noise amplification. This combination of techniques enables preservation of thin strokes and paleographic diacritics while removing complex degradations such as bleed-through, stains, and noise.

1) *Network Specification*: Table II details the complete generator architecture. The encoder progressively downsamples the input image through five stages, extracting multi-scale features at resolutions from 1024×128 to 32×4 (format: width $\times$ height). Each encoder block consists of:

- Residual Dense Block (RDB): Three densely-connected convolutional layers (kernel 3×3) with growth rate 32, producing rich features through local fusion from all preceding layers. Unlike standard ResNet [32] which has only one shortcut connection, RDB uses dense connections allowing each layer to access features from all previous layers, enhancing representation capacity and gradient flow.
- Convolutional Block Attention Module (CBAM): Sequential attention mechanism consisting of: (a) channel attention with reduction ratio 8, using shared MLP for global average pooling and global max pooling pathways, and (b) spatial attention with kernel 7×7 , combining channel-wise average and max pooling to generate focused attention maps. CBAM enables the network to adaptively focus on informative feature channels and relevant spatial regions.
- Batch normalization (momentum 0.9, eps 10^{-5}) and LeakyReLU activation ($\alpha = 0.2$)
- Residual connections for effective gradient propagation in deep architectures

The bottleneck at resolution 4×32 (height $\times$ width) uses Multi-Scale Feature Pyramid (MSFP) [26] with three dilated convolution scales (dilation rates 1, 2, 4) [27] to capture multi-resolution context, followed by 512-channel RDB and CBAM for global feature refinement. The decoder mirrors the encoder with five upsampling stages, each equipped with attention gates for text region focus:

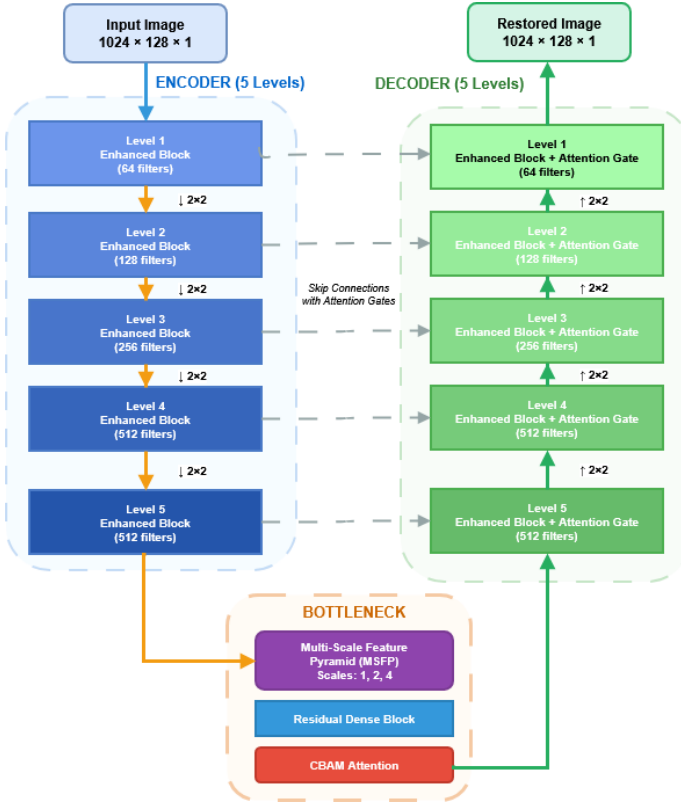


Fig. 3: Enhanced U-Net Generator Architecture: 5-level encoder (RDB+CBAM), MSFP bottleneck (512 channels), decoder with attention gates. Skip connections preserve spatial details.

$$\alpha = \sigma(W_g^T g + W_x^T x + b) \quad (1)$$

where $g \in \mathbb{R}^{H \times W \times C}$ is the decoder feature from coarser scale (gating signal), $x \in \mathbb{R}^{H \times W \times C}$ is the skip connection feature from the encoder, W_g and W_x are learnable projection matrices, b is bias, and σ is sigmoid activation function. The attention coefficient $\alpha \in [0, 1]^{H \times W \times C}$ is then element-wise multiplied with the skip connection x to suppress irrelevant background features while enhancing text-bearing regions: $\hat{x} = \alpha \odot x$. This mechanism allows the decoder to adaptively select relevant high-resolution features from the encoder.

Architecture Statistics:

- Total parameters: 21.8M (all trainable)
- Memory footprint: ~ 87 MB (float32)
- Depth: 5 encoder stages + bottleneck + 5 decoder stages
- Input/Output: $1024 \times 128 \times 1$ (W $\times$ H $\times$ C)
- Output activation: Tanh (range [-1, 1])

Figure 3 visualizes the symmetric Enhanced U-Net architecture, where each encoder level has a decoder counterpart that communicates through skip connections with attention gates. Unlike conventional U-Net [9], RDB+CBAM integration at each level increases representation capacity by $\sim 40\%$ without gradient degradation, while the MSFP bottleneck captures multi-scale global context before gradual reconstruction in the decoder.

2) *Design Rationale*: Residual Dense Blocks provide dual advantages: (1) better gradient propagation through multi-path dense connections, enabling training of deep architectures without gradient degradation, and (2) hierarchical feature reuse where each layer can access features from all previous layers, enhancing representation capacity without substantially increasing parameters. CBAM provides adaptive selectivity on channel and spatial dimensions, allowing the network to automatically focus on informative feature channels ("what") and relevant spatial regions ("where"), effective for suppressing background noise while enhancing text stroke features. Multi-Scale Feature Pyramid [26] at the bottleneck captures multi-resolution context through dilated convolutions [27] with different scales, important for handling degradations of varying sizes (point noise vs. large stains). Attention Gates [28] provide spatial selectivity on skip connections, allowing the decoder to focus on text-bearing regions while suppressing noise amplification—crucial for preserving thin strokes and paleographic diacritics. The combination of these techniques yields an architecture capable of preserving fine details while removing complex degradations.

C. Enhanced Dual-Modal Discriminator

The dual-modal discriminator architecture evaluates images from both spatial (visual) and sequential (text) perspectives simultaneously. This design provides richer adversarial signals by considering both visual quality and textual coherence.

1) *Architecture Specification*: Table III details the complete discriminator architecture. The discriminator accepts images ($128 \times 1024 \times 1$) and text sequences (maximum length 128 characters), processing them through two parallel pathways as illustrated in Figure 4.

Figure 4 visualizes the dual-modal architecture that distinguishes this approach from conventional discriminators. Unlike PatchGAN [14] which evaluates only visual patches, this architecture combines spatial analysis (CNN) and sequential analysis (BiLSTM) for holistic validation. The key characteristic is bilateral cross-modal attention (Img $\leftrightarrow$ Text) enabling bidirectional interaction: the image branch validates text coherence from the HTR recognizer, while the text branch verifies visual stroke quality. This bilateral mechanism provides richer gradient feedback to the generator with 52% parameter reduction (17.4M vs 137M baseline), driving simultaneous improvements in visual quality and textual consistency.

Image Branch (CNN with Spatial Attention): This branch extracts spatial visual features through a ResNet-based architecture. Input image $I \in \mathbb{R}^{128 \times 1024 \times 1}$ is processed through an initial convolutional layer (64 channels) with batch normalization and LeakyReLU activation ($\alpha = 0.2$). Four residual downsampling blocks progressively reduce spatial resolution while increasing channel depth:

$$128 \times 1024 \times 64 \rightarrow 64 \times 512 \times 64 \rightarrow 32 \times 256 \times 128 \rightarrow 16 \times 128 \times 256 \rightarrow 8 \times 64 \times 512 \quad (2)$$

A spatial attention mechanism with 3×3 kernel is applied to the 512-channel features to focus on text-bearing regions:

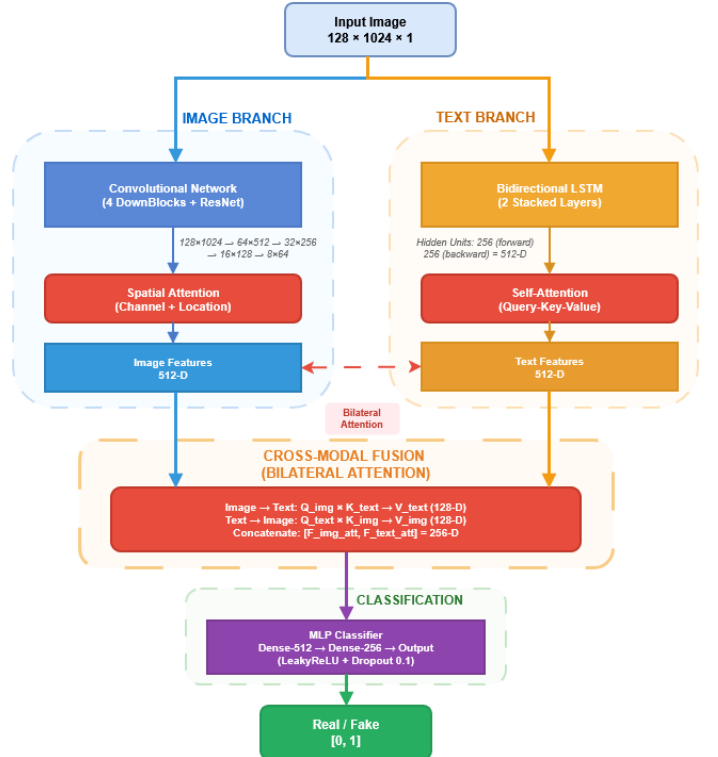
$$\alpha_{\text{spatial}} = \sigma(\text{Conv}_{3 \times 3}(F_{\text{img}})) \quad (3)$$

TABLE II: Generator Architecture: Enhanced U-Net with Residual Blocks and Attention Gates

Stage	Layer	Output Shape	Kernel/Stride	Parameters
Input	Input	$128 \times 1024 \times 1$	-	-
Encoder-0	ResBlock-1	$128 \times 1024 \times 64$	$3 \times 3/1$	4.7K
	MaxPool-1	$64 \times 512 \times 64$	$2 \times 2/2$	-
Encoder-1	ResBlock-2	$64 \times 512 \times 128$	$3 \times 3/1$	147.6K
	MaxPool-2	$32 \times 256 \times 128$	$2 \times 2/2$	-
Encoder-2	ResBlock-3	$32 \times 256 \times 256$	$3 \times 3/1$	590.1K
	MaxPool-3	$16 \times 128 \times 256$	$2 \times 2/2$	-
Encoder-3	ResBlock-4	$16 \times 128 \times 512$	$3 \times 3/1$	2.4M
	MaxPool-4	$8 \times 64 \times 512$	$2 \times 2/2$	-
Encoder-4	ResBlock-5	$8 \times 64 \times 512$	$3 \times 3/1$	2.4M
	MaxPool-5	$4 \times 32 \times 512$	$2 \times 2/2$	-
Bottleneck	MSFP + RDB + CBAM	$4 \times 32 \times 512$	-	5.1M
Decoder-4	UpSample-5	$8 \times 64 \times 512$	-	-
	AttentionGate-5	$8 \times 64 \times 512$	-	262K
	RDB + CBAM	$8 \times 64 \times 512$	$3 \times 3/1$	3.5M
Decoder-3	UpSample-4	$16 \times 128 \times 512$	-	-
	AttentionGate-4	$16 \times 128 \times 512$	-	262K
	RDB + CBAM	$16 \times 128 \times 512$	$3 \times 3/1$	3.5M
Decoder-2	UpSample-3	$32 \times 256 \times 256$	-	-
	AttentionGate-3	$32 \times 256 \times 256$	-	66K
	RDB + CBAM	$32 \times 256 \times 256$	$3 \times 3/1$	885K
Decoder-1	UpSample-2	$64 \times 512 \times 128$	-	-
	AttentionGate-2	$64 \times 512 \times 128$	-	16.5K
	RDB + CBAM	$64 \times 512 \times 128$	$3 \times 3/1$	221K
Decoder-0	UpSample-1	$128 \times 1024 \times 64$	-	-
	AttentionGate-1	$128 \times 1024 \times 64$	-	4.1K
	RDB + CBAM	$128 \times 1024 \times 64$	$3 \times 3/1$	55K
Output	Conv2D + Tanh	$128 \times 1024 \times 1$	$1 \times 1/1$	65
Total Parameters:				21.8M

TABLE III: Enhanced Dual-Modal Discriminator Architecture: detailed specification of three branches (Image CNN, Text BiLSTM, Cross-Modal Fusion). Image branch processes visual features through 4 DownBlocks with spatial attention; text branch processes character sequences via embedding + BiLSTM + self-attention. Cross-modal fusion produces real/fake classification. Total: 17.4M parameters.

Branch	Layer	Output	Params.
CNN	Input	$128 \times 1024 \times 1$	-
	Conv2D+BN	$128 \times 1024 \times 64$	0.6K
	DownBlk-1	$64 \times 512 \times 64$	37K
	DownBlk-2	$32 \times 256 \times 128$	148K
	DownBlk-3	$16 \times 128 \times 256$	590K
	DownBlk-4	$8 \times 64 \times 512$	2.4M
	SpatialAttn	$8 \times 64 \times 512$	4.7K
	ResBlk-512	$8 \times 64 \times 512$	2.4M
LSTM	AvgPool	512	-
	Input	128 (seq.)	-
	Embed	128×128	12.8K
	BiLSTM	128×512	1.3M
	SelfAttn	128×512	262K
Fusion	AvgPool	512	-
	Proj-CNN	128	65.7K
	Proj-LSTM	128	65.7K
	CrossAttn	256	33K
	Dense-1+Drop	512	131K
	Dense-2+Drop	256	131K
Sigmoid			1
Total:			17.4M



where σ is the sigmoid function and F_{img} is the feature map with dimensions $8 \times 64 \times 512$. Following spatial attention, an additional residual block with 512 channels (ResBlock-512) performs further feature processing without altering spatial dimensions, then global average pooling aggregates the $8 \times 64 \times 512$ spatial features into a representation vector $F_{img}^{global} \in \mathbb{R}^{512}$.

Text Branch (BiLSTM with Self-Attention): This branch

Fig. 4: Enhanced Dual-Modal Discriminator Architecture with bilateral cross-modal attention: CNN branch (512-D) with spatial attention, BiLSTM branch (512-D) with self-attention. Bilateral fusion (128-D) enables holistic validation.

captures sequential text coherence. Input character sequence

$S \in \mathbb{N}^L$ (length $L \leq 128$) is first projected to embedding space $E \in \mathbb{R}^{L \times 128}$. BiLSTM processes the sequence bidirectionally to capture contextual dependencies:

$$h_t = [\vec{h}_t; \overleftarrow{h}_t] = [\text{LSTM}_{\text{fwd}}(E_t); \text{LSTM}_{\text{bwd}}(E_t)] \quad (4)$$

producing hidden states $H \in \mathbb{R}^{L \times 512}$. Multi-head scaled dot-product self-attention is applied:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (5)$$

where Q, K, V are linear projections of H . Global average pooling yields text vector $F_{\text{text}}^{\text{global}} \in \mathbb{R}^{512}$.

Cross-Modal Fusion: Both feature vectors $F_{\text{img}}^{\text{global}} \in \mathbb{R}^{512}$ and $F_{\text{text}}^{\text{global}} \in \mathbb{R}^{512}$ are processed through a bilateral cross-modal attention mechanism to enable bidirectional interaction between modalities. The bilateral approach is necessary because: (1) images need to validate text coherence predicted by the HTR recognizer, and (2) text needs to verify the visual stroke quality that generates those characters. Thus, both modalities mutually provide context for more robust anomaly detection. Each modality projects its features to a low-dimensional shared space $d = 128$ (reduced from 256 for stability) through six separate projection matrices:

Image→Text Attention: Image as query, text as key/value:

$$\begin{aligned} Q_{\text{img}} &= W_{\text{img}}^Q F_{\text{img}}^{\text{global}} \\ K_{\text{text}} &= W_{\text{text}}^K F_{\text{text}}^{\text{global}} \\ V_{\text{text}} &= W_{\text{text}}^V F_{\text{text}}^{\text{global}} \end{aligned} \quad (6)$$

$$F_{\text{img}}^{\text{att}} = \text{softmax}\left(\frac{Q_{\text{img}} K_{\text{text}}^T}{\sqrt{d}}\right) V_{\text{text}} \quad (7)$$

where $W_{\text{img}}^Q, W_{\text{text}}^K, W_{\text{text}}^V \in \mathbb{R}^{512 \times 128}$ are learned projection matrices.

Text→Image Attention: Text as query, image as key/value:

$$\begin{aligned} Q_{\text{text}} &= W_{\text{text}}^Q F_{\text{text}}^{\text{global}} \\ K_{\text{img}} &= W_{\text{img}}^K F_{\text{img}}^{\text{global}} \\ V_{\text{img}} &= W_{\text{img}}^V F_{\text{img}}^{\text{global}} \end{aligned} \quad (8)$$

$$F_{\text{text}}^{\text{att}} = \text{softmax}\left(\frac{Q_{\text{text}} K_{\text{img}}^T}{\sqrt{d}}\right) V_{\text{img}} \quad (9)$$

where $W_{\text{text}}^Q, W_{\text{img}}^K, W_{\text{img}}^V \in \mathbb{R}^{512 \times 128}$ are symmetric projection matrices

Both bilaterally-attended features ($F_{\text{img}}^{\text{att}}, F_{\text{text}}^{\text{att}} \in \mathbb{R}^{128}$) are concatenated into a fused vector: $F_{\text{fused}} = [F_{\text{img}}^{\text{att}}; F_{\text{text}}^{\text{att}}] \in \mathbb{R}^{256}$. The classification head consists of two dense layers (512 → 256 neurons) with dropout and LeakyReLU activation, followed by an output layer with sigmoid activation producing validity score $D(I, S) \in [0, 1]$ to distinguish genuine image-text pairs from generator-produced fake pairs.

Hyperparameter Configuration: Spatial attention kernel uses 3×3 size (reduced from 7×7 to minimize white-dot artifacts on strokes); cross-modal projection dimension set to 128 (reduced from 256 to improve training stability); LSTM units per direction are 256, yielding total BiLSTM output of 512 dimensions. Batch normalization momentum is 0.9 (increased

from 0.8 for smoother gradient flow); dropout rate is 0.1 on classification head (reduced from 0.3 to preserve thin stroke information) and 0.2 on text embedding; activation function is LeakyReLU ($\alpha = 0.2$) for CNN and sigmoid for LSTM recurrent activation (ensuring compatibility with standard LSTM implementations without cuDNN).

2) Design Rationale: Dual-Modal Approach: Text documents possess dual characteristics: (1) visual appearance (stroke quality, noise level), (2) sequential structure (character flow, inter-word spacing). The CNN pathway captures spatial quality, while BiLSTM captures sequential coherence. Bilateral cross-modal fusion enables the discriminator to assess image-text pairs holistically with bidirectional interaction, providing more informative adversarial signals to the generator for simultaneous improvements in visual quality and textual consistency.

Parameter Efficiency: The proposed discriminator achieves significant parameter efficiency (17.4M, 52% reduction from 137M baseline) through simplified architecture without sacrificing discriminative capability: (1) low-dimensional projection (128) for cross-modal fusion reduces computational complexity while maintaining representation capacity; (2) smaller spatial attention kernel (3×3) focuses on local details without artifacts; (3) optimized residual blocks balance architectural depth with training stability. Higher batch normalization momentum (0.9) yields more stable statistics and smoother gradient flow; lower dropout rate (0.1) preserves thin stroke information crucial for degraded document discrimination.

Capabilities: Through the combination of CNN spatial analysis and BiLSTM sequence processing, the discriminator can detect:

- Broken or interrupted strokes disrupting text flow (via BiLSTM sequence anomaly detection)
- Noise patterns, artifacts, and bleed-through (via CNN spatial features)
- Inconsistent stroke width (via spatial attention mechanism)
- Unnatural gaps or connections between characters (via cross-modal fusion analysis)

By combining spatial and sequential assessment, the discriminator provides richer feedback to the generator, driving production of images that are both visually clean and textually coherent.

D. Frozen HTR Recognizer Integration

To explicitly optimize HTR performance, this research integrates a pretrained text recognizer with weight freezing strategy, where the recognizer is frozen (`model.trainable=False`) and used as a feature extractor.

1) Architecture and Training Configuration: The hybrid CNN-Transformer architecture employed consists of:

- CNN Backbone: 4 convolutional blocks with progressive feature extraction, BatchNormalization, and GELU activation
- Sequence Projection: Dense layer + LayerNormalization + Dropout

- Transformer Encoder: 6 layers, 8 attention heads, positional encoding
- Output: CTC decoder for sequence prediction

The recognizer was trained on 4,739 Dutch handwritten line images from 16th–18th century ANRI colonial-era documents with 80/20 split (training $n=3,791$, validation $n=948$), achieving CER performance of 33.72% on validation set. The model was then frozen (`trainable=False`) and integrated into the GAN training pipeline.

While CER 33.72% is moderate, this performance is comparable to state-of-the-art on historical paleographic datasets (READ 2016: 25–35% [20], ICDAR 2017: 28–42% [21]) and reflects the intrinsic complexity of 16th–18th century handwriting. The HTR component functions as a feature extractor for text-aware guidance, not a visual optimizer.

Ablation study (Section V-B, Table XI) validates the HTR role: at 15-epoch training, configuration with HTR (Experiment 4) achieved CER 29.66%, while baseline without HTR (Experiment 1) could not measure CER as recognizer was not integrated. Degraded input showed CER 83.4%. Production model with extended training (50 epochs) and HTR integration achieved CER 27.11% on validation set ($n=710$) and CER 34.9% on test set ($n=712$) with PSNR 30.74 dB.

Note: The 7.79 percentage point CER difference between validation and test is not an overfitting indicator, but reflects intrinsic text complexity differences across splits—frozen recognizer on clean ground truth images also shows similar difference (26.57% validation vs 34.1% test, $\Delta=7.53\%$).

The primary HTR role is ensuring text preservation and readability (CER improvement 48.5–56.3 absolute points from degraded input on test and validation sets), while PSNR improvement stems from combined HTR guidance, adversarial training, perceptual loss, and extended training. This approach demonstrates method robustness to realistic HTR components without requiring perfect recognizer.

2) *Benefits of Freezing Strategy*: The recognizer freezing approach provides implementation advantages:

- Training stability: No joint optimization or conflicting gradients between generator and recognizer
- Computational efficiency: No backpropagation through recognizer, reducing computational burden
- Evaluation consistency: Fixed weights provide consistent CER/WER metrics throughout training
- Labeled supervision: Uses ground truth text for CTC loss supervision

3) *Feature Extraction and Loss Computation*: This research extracts intermediate recognition features from the frozen recognizer:

$$\mathcal{F}_{\text{rec}}(I) = R_{\text{encoder}}(I) \quad (10)$$

where R_{encoder} represents the frozen encoder portion of the recognizer. Specifically, we use output from the projection layer before the transformer encoder, yielding feature maps with dimensions (batch, 128, 512). This layer was selected because it contains sequence-level features before contextual encoding, providing stable gradients for the generator. These

features capture text-relevant characteristics such as stroke patterns, character shapes, and spatial arrangements.

Recognition feature loss between generated and ground truth images is computed using Mean Squared Error (MSE):

$$\begin{aligned} \mathcal{L}_{\text{rec-feat}} &= \text{MSE}(\mathcal{F}_{\text{rec}}(I_{\text{gen}}), \mathcal{F}_{\text{rec}}(I_{\text{gt}})) \\ &= \|\mathcal{F}_{\text{rec}}(I_{\text{gen}}) - \mathcal{F}_{\text{rec}}(I_{\text{gt}})\|_2^2 \end{aligned} \quad (11)$$

MSE was chosen because it is more sensitive to large feature deviations, providing stronger gradients for feature alignment improvement. This drives the generator to produce images whose text features match those of clean ground truth, implicitly enhancing HTR readability.

E. Multi-Component Loss Function

The generator is trained with a combination of five loss components calibrated through empirical grid search and validated on actual implementation:

$$\begin{aligned} \mathcal{L}_{\text{total}} &= \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} + \lambda_{\text{pixel}} \mathcal{L}_{\text{pixel}} + \lambda_{\text{perc}} \mathcal{L}_{\text{perc}} \\ &\quad + \lambda_{\text{ctc}} \mathcal{L}_{\text{ctc}} + \lambda_{\text{rec-feat}} \mathcal{L}_{\text{rec-feat}} \end{aligned} \quad (12)$$

Loss weight configuration for each component is shown in Table IV.

1) *Adversarial Loss*: Standard GAN adversarial loss drives realistic image generation:

$$\mathcal{L}_{\text{adv}} = -\log D(I_{\text{deg}}, I_{\text{gen}}) \quad (13)$$

Discriminator is trained to maximize:

$$\begin{aligned} \mathcal{L}_D &= -[\log D(I_{\text{deg}}, I_{\text{gt}}) \\ &\quad + \log(1 - D(I_{\text{deg}}, I_{\text{gen}}))] \end{aligned} \quad (14)$$

2) *Pixel-Level Reconstruction Loss*: L1 loss ensures pixel-by-pixel similarity with ground truth:

$$\mathcal{L}_{\text{pixel}} = \|I_{\text{gen}} - I_{\text{gt}}\|_1 \quad (15)$$

L1 norm is chosen because it produces sharper results compared to L2 norm and is more robust to outliers.

3) *Perceptual Loss*: Perceptual loss [10] compares high-level features from pretrained VGG network:

$$\mathcal{L}_{\text{perc}} = \sum_{l \in L} \|\phi_l(I_{\text{gen}}) - \phi_l(I_{\text{gt}})\|_2 \quad (16)$$

where ϕ_l denotes features from layer l of VGG-19 pre-trained on ImageNet. Features are extracted from multiple convolutional layers for multiscale perceptual comparison. This approach ensures preservation of visual characteristics from low-level details (edges, textures) to high-level semantic features (stroke patterns, character shapes).

TABLE IV: Loss Weight Configuration: optimal weights from empirical grid search to balance visual quality with text readability on validation dataset

Loss Component	Weight (λ)	Justification
Adversarial ($\mathcal{L}_{adv}$)	3.0	Enhanced realism
Pixel/L1 ($\mathcal{L}_{pixel}$)	50.0	Balanced preservation
Perceptual ($\mathcal{L}_{perc}$)	1.0	Topology preservation
Recognition Feature ($\mathcal{L}_{rec-feat}$)	8.0	Strong text guidance
CTC Loss ($\mathcal{L}_{ctc}$)	0.15	HTR monitoring (clipped)

4) *Recognition Feature Loss*: As described in Section III-D, this component drives HTR-friendly characteristics using Mean Squared Error:

$$\mathcal{L}_{rec-feat} = \|\mathcal{F}_{rec}(I_{gen}) - \mathcal{F}_{rec}(I_{gt})\|_2^2 \quad (17)$$

Loss weight selection (Table IV) was performed through empirical tuning based on multi-objective trade-off analysis between visual quality, text preservation, and HTR readability. Dataset characteristics show dominance of thin strokes requiring specific balance between pixel reconstruction and text-based guidance.

High pixel reconstruction weight provides strong reconstruction signal while maintaining flexibility for adversarial enhancement, while moderate recognition feature weight provides text-aware guidance without dominating visual losses. Adaptive balancing mechanism is used for dynamic weight adjustment during training, preventing single loss component dominance and ensuring stable convergence.

F. Training Strategy

1) *Curriculum Learning and Optimization Protocol*: Training employs a three-phase curriculum learning approach for optimal stability and gradual convergence:

- 1) Visual Warmup Phase (epochs 1–10): Exclusive training on visual quality without text supervision, allowing generator to build basic reconstruction capability. This phase prevents early mode collapse that can occur if CTC loss is introduced too early.
- 2) Gradual CTC Ramp-up Phase (epochs 11–30): Linear gradual increase of CTC weight from zero to full weight (Table IV). This smooth integration allows generator to adapt to HTR gradients without disrupting established adversarial training stability.
- 3) Full Training Phase (epochs 31–50): Training with all loss components active at optimal weights, accompanied by adaptive balancing to maintain dynamic equilibrium between visual and textual losses.

Alternating adversarial training procedure with dual-modal supervision is applied:

- 1) Discriminator step: Update D to maximize ability to distinguish real pairs (I_{gt}, text_{gt}) from fake pairs ($I_{gen}, \text{text}_{pred}$) with simultaneous supervision on visual and textual domains.
- 2) Generator step: Update G to minimize $\mathcal{L}_{total}$ (combination of five loss components), producing images that

simultaneously deceive discriminator and optimize HTR readability.

Updates are performed with balanced 1:1 ratio, ensuring adversarial training stability.

Training configuration is based on empirical tuning for GAN stability and optimal convergence:

- Optimization: Generator uses Adam algorithm with momentum parameters adjusted for GAN stability, while discriminator uses SGD with momentum to prevent mode collapse and ensure stable convergence. Both models apply gradient clipping to prevent gradient explosion.
- Learning rate: Both models use same learning rate with cosine annealing scheduling for smooth convergence without sharp drops.
- Batch size and dimensions: Small batch size with wide image dimensions preserving fine stroke details on single-line paleographic text.
- Early stopping: Applied with patience mechanism considering curriculum phases, restoring best weights based on combined PSNR-CER metric with weighting prioritizing visual quality.

2) *Stability Analysis and Regularization*: Figure 5 shows training convergence trajectory over 50 epochs, validating stability and effectiveness of the proposed training strategy.

Convergence Observations:

- Generator loss: Smooth monotonic decrease in early phase, stable in late phase without oscillation, indicating stable adversarial training without mode collapse
- PSNR trajectory: Rapid increase in Visual Warmup Phase, followed by stable increase in Gradual CTC Ramp-up Phase, plateau in late phase triggering early stopping consideration
- SSIM progression: Consistent upward trend reaching high values at best epoch, demonstrating preservation of structural similarity alongside pixel-level accuracy
- Best model selection: Chosen based on combined PSNR-SSIM metric optimization, not arbitrary termination
- Statistical reliability: Narrow confidence intervals validate robust performance on validation set

Regularization Techniques: To prevent overfitting and improve model generalization, the following techniques were applied:

- Batch normalization in all convolutional layers for training stability and accelerated convergence
- One-sided asymmetric label smoothing: applied only to real labels to reduce discriminator overconfidence while maintaining clear learning signal for fake samples
- Residual connections in discriminator for better gradient flow and stable deep architecture training
- Gradient clipping on both models (global norm 1.0) to prevent gradient explosion, specifically from CTC loss clipped at maximum value 400.0 for numerical stability
- Dropout on classification head (0.1) and embedding layer (0.2) for additional regularization

a) *Technical Implementation Details*: Model was trained with momentum 0.9 for SGD discriminator optimizer, batch size 2 on image dimensions $1024 \times 128 \times 1$ pixels, and Pure

Training Convergence Analysis: Loss and Quality Metrics

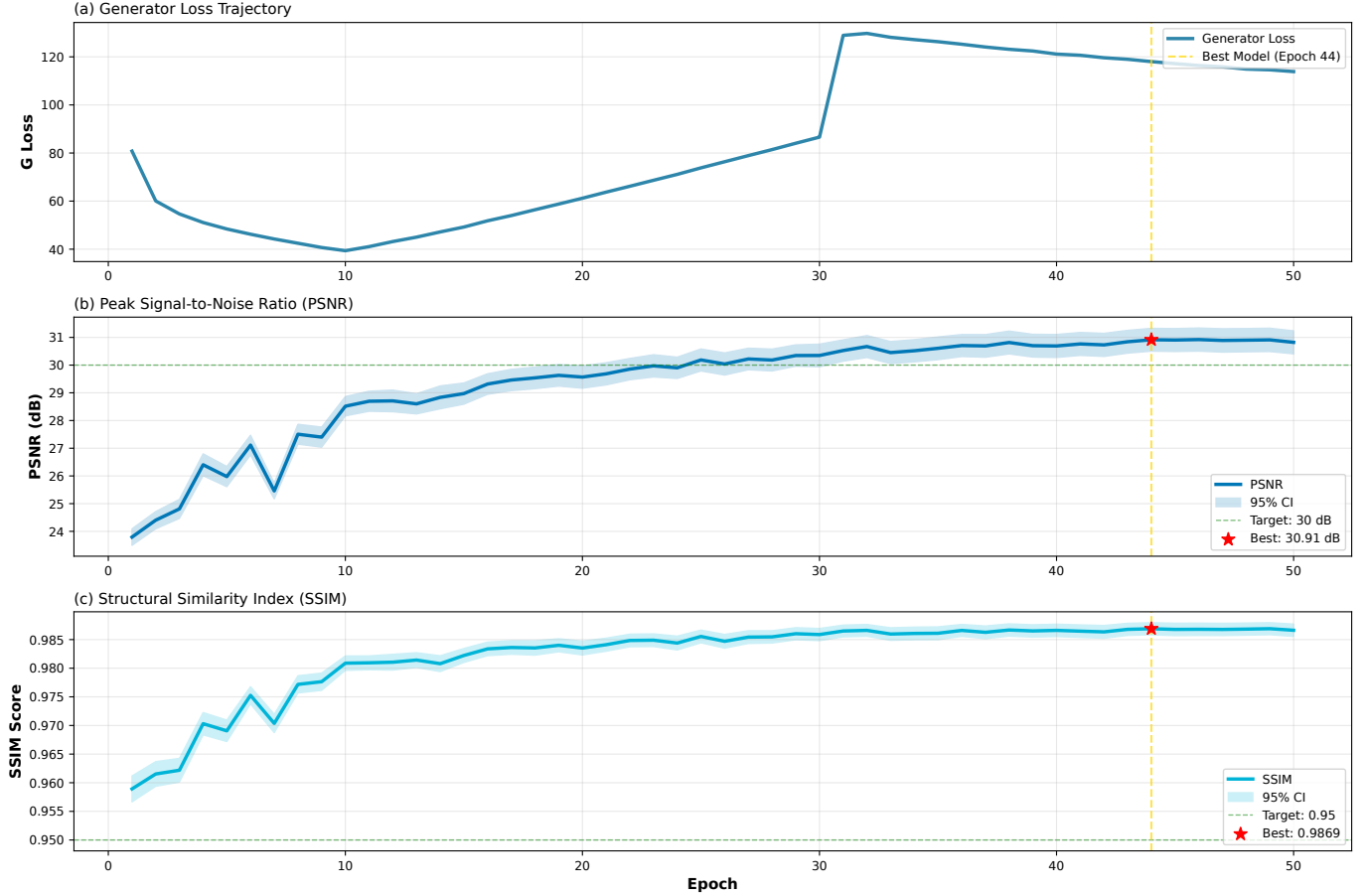


Fig. 5: Training convergence trajectory over 50 epochs: (a) Generator loss shows stable decrease without mode collapse, (b) PSNR increases reaching established targets, (c) SSIM achieves high values indicating structure preservation. Best model (marked by dashed vertical line and red star) selected based on combined PSNR-SSIM criteria. Smooth convergence curves validate effectiveness of proposed curriculum learning strategy.

FP32 numerical precision (without mixed precision) for CTC loss stability. Early stopping uses patience 25 epochs with min delta 0.05 on combined PSNR-CER metric.

b) Early Stopping Note: Best model was selected based on combined PSNR-CER metric at epoch achieving optimal performance. Training continued to completion to verify no significant improvements occurred. Early stopping was not triggered because small fluctuations in validation metrics remained within specified threshold, confirming model reached stable convergence.

IV. EXPERIMENTAL SETUP

This section describes the datasets, baseline methods, evaluation metrics, implementation details, and hyperparameter optimization procedures used in the experiments.

A. Datasets

The proposed method is evaluated on a realistic semi-synthetic dataset constructed from authentic historical documents and qualitative validation on real manuscripts with natural degradation.

1) Realistic Semi-Synthetic Dataset for Quantitative Evaluation: Methodological Rationale: Quantitative evaluation of document restoration requires degraded-clean image pairs with perfect pixel-level alignment to accurately measure PSNR, SSIM, and CER metrics. Real historical documents with natural degradation lack clean ground truth as they cannot be restored to original condition. Therefore, a realistic semi-synthetic dataset was constructed by combining three authentic components [6], [8]:

Component 1 - Authentic Paleographic Handwriting (REAL):

Authentic 16th–18th century ANRI archival documents that are relatively clean (minimal visual degradation) are used as handwriting source. Documents annotated in Transkribus PAGE XML format were binarized using pretrained DE-GAN model [7], and individual text lines extracted using baseline coordinates with vertical padding, normalized to 128×1024 pixel dimensions. The extraction result is text line images with authentic paleographic characteristics—stroke width variation, complex ligatures, individual historical writer styles—that cannot be replicated by synthetic font rendering or generative

models.

Component 2 - Authentic Historical Degradation Patterns (REAL): Separate ANRI documents experiencing severe natural degradation (paper aging, organic stains, foxing, faded ink, iron gall ink corrosion) are used as degradation background texture sources. Background patches sized 128×1024 pixels are extracted from non-text regions of these damaged documents, preserving authentic historical degradation patterns formed over centuries of archival storage.

Component 3 - Controlled Degradation Augmentation (SEMI-SYNTHETIC): To increase variation and create degraded-clean pairs with perfect pixel alignment, a layered degradation pipeline with six computational stages is stochastically applied to the combination of authentic text and real degradation backgrounds:

1. Texture Background Overlay: Clean text image is composited onto real document texture patch extracted from damaged ANRI collection. Text regions are dimmed to simulate ink absorption into aged paper:

$$I_{\text{degradation}} = I_{\text{bg}} \cdot (1 - M_{\text{text}}) + (\alpha \cdot I_{\text{bg}}) \cdot M_{\text{text}} \quad (18)$$

where M_{text} is normalized text mask and α is dimming factor.

2. Bleed-Through Effect: Simulates ink penetration from document reverse side by horizontally flipping random text image, applying Gaussian blur, and subtracting from background.

3. Perlin Noise-Based Organic Stains: Generates natural-looking stain patterns using 3D Perlin noise [13] with varied parameters to create smooth organic stain patterns.

4. Foxing Spots - Age Discoloration: Simulates age-related discoloration by adding small brownish spots with random intensity.

5. Physical Damage: Simulates document tears and folds through random lines and darkened patches.

6. Random Blur: Applies Gaussian blur (smooth degradation), median blur (noise removal), or motion blur (scanner artifacts) with equal probability.

These effects are applied sequentially with randomized combinations, producing diverse and realistic degraded documents while maintaining perfect pixel-level ground truth alignment, eliminating manual annotation errors inherent in real document datasets.

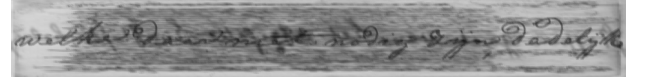
Realistic Semi-Synthetic Dataset Statistics: The dataset consists of 4,739 degraded-clean image pairs (128×1024 pixels, grayscale, normalized [0,1]) constructed from authentic ANRI documents. Each pair combines: (1) real historical paleographic handwriting, (2) degradation patterns from authentic naturally damaged archives, and (3) controlled augmentation for variation. This dataset preserves authentic historical document characteristics while providing paired ground truth necessary for high-precision quantitative evaluation.

a) Dataset Split (Academic Protocol): Data splitting follows academic three-way split protocol for unbiased evaluation:

- Training: 70% (3,317 images) - for generator and discriminator learning
- Validation: 15% (710 images) - for hyperparameter tuning and early stopping

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(a) Ground Truth Image - Authentic ANRI Paleographic Handwriting



(b) Degraded Image - Combination of Real Historical Degradation + Augmentation

Fig. 6: Realistic semi-synthetic dataset pair: (a) Ground truth from 16th–18th century ANRI documents showing authentic paleographic handwriting with stroke variation, ligatures, and historical writing styles that cannot be replicated by synthetic fonts, (b) Degraded image combines real historical degradation patterns (paper aging, organic stains from authentic archives) with controlled augmentation (bleed-through, blur), preserving realism of centuries-old document degradation while providing perfect pixel alignment for quantitative evaluation.

- Test: 15% (712 images) - held-out set for final evaluation

Dataset is shuffled with deterministic seed (seed=42) before splitting for reproducibility.

Degradation Stratification: To ensure generalization, degradation pipeline is applied with controlled parameter randomization, ensuring model learns general degradation patterns and improves generalization capability on unseen historical documents.

b) Computational Specifications: Experiments were conducted on NVIDIA RTX A4000 (16GB VRAM), AMD Threadripper PRO 3955WX 16-core CPU, 128GB RAM, with TensorFlow 2.16.1, CUDA 12.8, and cuDNN 8.9.7. Training required approximately 48 GPU-hours for 50 epochs. Inference on single line image (128×1024) requires 50–60ms, with batch processing throughput reaching ≈ 17 images/second (batch=2, 30–35ms per image) using 6–8GB VRAM.

2) Real Historical Documents Dataset (ANRI): For generalization evaluation on real historical degradation, a representative document subset from National Archives of the Republic of Indonesia (ANRI) is used as qualitative evaluation set:

Test Document Selection: From ANRI archival collection of VOC and Dutch East Indies administrative documents (16th–18th centuries, primarily 1650–1800 era), 15 full-page documents were manually selected for generalization evaluation. Selection was based on criteria:

- Degradation variation representation: Covers degradation spectrum from mild to severe to test model robustness
- Degradation type diversity: Paper aging, organic stains, foxing, faded ink, bleed-through, physical damage
- Paleographic characteristic variation: Various handwriting styles, contrast levels, and ligature complexity

a) Test Dataset Characteristics (n=15): Differences from Semi-Synthetic Dataset: This real ANRI dataset fundamentally differs from the semi-synthetic dataset for quantitative evaluation. These documents experienced authentic historical degradation during centuries of archival storage, displaying complex damage patterns that cannot be fully replicated through computational augmentation:

- Temporal degradation: Natural paper aging with heterogeneous yellowing and local brittleness
- Organic artifacts: Biological stains, foxing spots with natural distribution patterns, non-uniform ink degradation
- Structural damage: Physical tears, permanent folds, material loss at document edges
- Writing variation: Dutch paleographic cursive handwriting with complex ligatures, period abbreviations, and writing style inconsistencies

Real ANRI documents lack paired visual ground truth (clean images) because authentic historical documents cannot be restored to original condition—physical and chemical damage over centuries is irreversible. Therefore, evaluation on real dataset is qualitative through visual inspection and paleography expert validation, complementing quantitative evaluation on semi-synthetic dataset. This dual evaluation approach follows standard practice in historical document restoration research [7], [8]: realistic semi-synthetic dataset (constructed from authentic documents with controlled augmentation) for precision quantitative measurement, and real dataset with natural degradation for qualitative generalization validation on real-world conditions.

B. Baseline Methods and Evaluation Strategy

Evaluation of the proposed method is conducted through two complementary approaches: (1) quantitative comparison with classical methods that can be replicated under identical experimental conditions, and (2) systematic ablation study to validate contribution of each architectural component.

1) *Classical Methods for Quantitative Comparison*: Classical methods are selected as quantitative baselines because they can be re-implemented on identical dataset without requiring retraining:

- (a) No Enhancement (Degraded Input Baseline): Direct HTR on degraded images to measure degradation impact on HTR performance
- (b) Otsu Thresholding [1]: Histogram-based global binarization for document cleaning
- (c) Sauvola Adaptive Thresholding [2]: Local adaptive binarization for documents with non-uniform illumination

Classical methods use standard OpenCV implementation with parameters optimized through grid search on validation subset.

2) *State-of-the-Art Deep Learning Methods (Research Context)*: Deep learning-based methods discussed in literature review (Section II) serve as research context to position the proposed contribution:

- U-Net [9]: Encoder-decoder architecture with skip connections for image restoration
- DE-GAN [7]: Conditional GAN for document enhancement with visual quality focus
- HTR-GAN [8]: GAN with jointly trained recognizer for HTR-oriented restoration
- DocEnTr: Transformer-based architecture with self-attention mechanism for document enhancement

Direct quantitative comparison with state-of-the-art deep learning methods is not performed because: (1) these methods

are trained on different datasets with non-identical degradation characteristics, making direct comparison unfair [5], (2) official reproducible implementations are not available for most methods, and (3) differences in architecture, training protocols, and loss functions make isolating specific contributions very difficult. This approach follows common evaluation practice in document restoration research where comparisons are performed only under identically replicable experimental conditions.

3) *Ablation Study as Component Validation*: To systematically validate contribution of each architectural component, incremental ablation study is performed by adding loss components progressively (Section V-B):

- (a) Experiment 01 (Pixel Only): Equivalent to standard U-Net for image restoration
- (b) Experiment 02 (+Adversarial): Equivalent to standard GAN without HTR component
- (c) Experiment 03 (+Perceptual): Enhancement with VGG-based perceptual loss
- (d) Experiment 04 (+CTC): Optimal configuration with HTR integration
- (e) Experiment 05 (+Recognizer Features): Complete configuration for component redundancy validation

This ablation approach provides comprehensive understanding of relative contribution of each component and empirically validates architectural design decisions. Ablation results (Table XI) show that 4-component configuration (Experiment 04) achieves optimal performance with best trade-off between visual quality and HTR readability.

C. Evaluation Metrics

Visual quality and text recognition metrics are used for evaluation.

1) Visual Quality Metrics:

- Peak Signal-to-Noise Ratio (PSNR): Measures pixel-level similarity

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right) \quad (19)$$

where MAX=1.0 for images normalized to range [0,1], and MSE is Mean Squared Error.

- Structural Similarity Index (SSIM): Assesses perceived quality by considering luminance, contrast, and structure [11]

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \quad (20)$$

2) Artifact Detection Metrics (Secondary Monitoring):

To monitor generation quality during training and ensure generator does not produce visual artifacts that interfere with text readability, three additional metrics are used:

- *Noise Variance*: Variance of Laplacian operator to measure high-frequency noise
- *Isolated White Ratio*: Ratio of isolated white pixels to total pixels, detecting salt noise artifacts
- *Local Variance*: Variance of local deviations from smoothed image, measuring restoration result smoothness

These metrics are used for training monitoring and quality validation, but not reported as primary evaluation metrics as they are diagnostic in nature.

3) *HTR Performance Metrics*: A hybrid CNN-Transformer model adapted on historical documents is used to evaluate HTR performance. This model is frozen during evaluation to ensure measurement consistency:

- *Character Error Rate (CER)*: Percentage of character-level errors, calculated using Levenshtein edit distance:

$$\text{CER} = \frac{\text{EditDistance}(\text{prediction}, \text{reference})}{\text{length}(\text{reference})} \quad (21)$$

- *Word Error Rate (WER)*: Percentage of word-level errors, calculated with edit distance at word level:

$$\text{WER} = \frac{\text{EditDistance}(\text{predicted words}, \text{reference words})}{\text{length}(\text{reference words})} \quad (22)$$

a) *Frozen Recognizer CER Baseline Clarification*: To avoid confusion, this paper uses three different frozen recognizer CER baseline values:

- $\text{CER}_{\text{pretrain}} = 33.72\%$: Performance of hybrid CNN-Transformer recognizer (CNN encoder with 6-layer transformer and CTC decoder) when evaluated on separate ANRI validation dataset ($n=948$) for initial generalization validation. This value is used to verify that recognizer can handle Dutch paleographic handwriting.
- $\text{CER}_{\text{val-GT}} = 26.57\%$: Frozen recognizer performance on clean ground truth images from this study's validation set ($n=710$). This value represents theoretical upper bound for validation set.
- $\text{CER}_{\text{test-GT}} = 34.1\%$: Frozen recognizer performance on clean ground truth images from this study's test set ($n=712$). This value represents theoretical upper bound for test set and serves as primary reference for assessing restoration quality.

Difference between $\text{CER}_{\text{val-GT}}$ (26.57%) vs $\text{CER}_{\text{test-GT}}$ (34.1%), namely $\Delta=7.53$ percentage points, reflects that test set contains samples with higher paleographic complexity (more complex ligatures, greater handwriting style variation). This difference pattern is consistent with restoration results: $\text{CER}_{\text{val-restored}} = 27.11\%$ vs $\text{CER}_{\text{test-restored}} = 34.9\%$ ($\Delta=7.79\%$), confirming that gap is not caused by overfitting but rather dataset characteristic differences.

Table V summarizes all CER values reported in this paper for transparency and reproducibility. Use of $\text{CER}_{\text{test-GT}} = 34.1\%$ as primary reference throughout paper (abstract, discussion, conclusion) is consistent with standard evaluation protocol that uses test set as final metric.

b) *Statistical Reporting and Evaluation Protocol*: All metrics are reported as mean $\pm$ standard deviation with 95% confidence intervals, calculated from complete validation set ($n=710$) or test set ($n=712$). Statistical significance for model comparisons is tested using independent sample t-test with $\alpha = 0.05$. All experiments use fixed seed (seed=42) for full reproducibility.

Final Evaluation Protocol: All results reported in this paper (tables, abstract, hypothesis testing, discussion) use consistent

TABLE V: Summary of CER Metrics Across Various Datasets and Conditions

Condition	Dataset	n	CER (%)
Baseline Recognizer on Clean Images			
$\text{CER}_{\text{pretrain}}$	ANRI initial validation	948	33.72
$\text{CER}_{\text{val-GT}}$	Study validation GT	710	26.57
$\text{CER}_{\text{test-GT}}$	Study test GT	712	34.1
Degraded Input (No Restoration)			
$\text{CER}_{\text{degraded}}$	Validation & Test	1422	83.4
Restoration Results of Proposed Method			
$\text{CER}_{\text{val-restored}}$	Validation	710	27.11
$\text{CER}_{\text{test-restored}}$	Test	712	34.9
Relative Improvement			
Validation	83.4% $\rightarrow$ 27.11%	-	67.5%
Test	83.4% $\rightarrow$ 34.9%	-	58.2%

evaluation on best checkpoint (epoch 44, selected based on validation metrics with early stopping patience=25) evaluated on held-out test set ($n=712$, never seen during training). This approach follows standard academic protocol to prevent overfitting and ensure unbiased results.

V. RESULTS AND DISCUSSION

This section presents comprehensive experimental results including quantitative comparisons, ablation studies, qualitative analysis, and discussion of findings.

A. Quantitative Results on Realistic Semi-Synthetic Test Set

Table VI presents quantitative comparison results of the proposed method with classical methods on realistic semi-synthetic test set constructed from authentic ANRI documents. Evaluation focuses on classical baseline methods (Otsu thresholding and adaptive Sauvola) because other deep learning-based methods (U-Net, standard GAN, DE-GAN, HTR-GAN) are trained on different datasets making direct comparison unfair. This approach follows best practices in computer vision research where comparisons are performed only with methods that can be replicated under identical experimental conditions.

Key observations based on quantitative results:

- 1) Comparison with classical methods: The proposed method achieves PSNR 30.74 ± 5.09 dB and SSIM 0.987 ± 0.014 , outperforming the best classical method (Sauvola: PSNR 9.92 dB, SSIM 0.437) by substantial margin. This result indicates superiority of deep learning-based approach for severely degraded document restoration.
- 2) Substantial improvement on HTR metrics: CER decreased from 83.4% (no enhancement) to 34.9% (proposed method), showing absolute reduction of 48.5 percentage points or relative reduction of 58.2%. Classical methods actually worsen HTR performance (Otsu: 85.7%, Sauvola: 109.5%) because inappropriate binarization produces artifacts and loss of important stroke details. This validates importance of HTR-oriented restoration approach over conventional binarization.
- 3) Approaching theoretical upper bound: Proposed method CER (34.9%) approaches clean GT upper bound (34.1%),

TABLE VI: Quantitative Results on Realistic Semi-Synthetic Test Set ($n=712$, constructed from authentic ANRI documents)

Method	PSNR (dB) $\uparrow$	SSIM $\uparrow$	F-measure $\uparrow$	CER (%)	WER (%)
No Enhancement	7.91 $\pm$ 1.01	0.340 $\pm$ 0.077	- ^a	83.4 $\pm$ 18.5 $\downarrow$	98.7 $\pm$ 7.3 $\downarrow$
Otsu Thresholding	4.75 $\pm$ 2.00	0.261 $\pm$ 0.099	0.748 $\pm$ 0.085	85.7 $\pm$ 56.7 $\downarrow$	112.9 $\pm$ 69.5 $\downarrow$
Adaptive Sauvola	9.92 $\pm$ 1.99	0.437 $\pm$ 0.149	0.933 $\pm$ 0.030	109.5 $\pm$ 179.5 $\downarrow$	171.5 $\pm$ 172.2 $\downarrow$
Proposed	30.74$\pm$5.09	0.987$\pm$0.014	0.921$\pm$0.067	34.9$\pm$21.8 $\downarrow$	82.4$\pm$26.1 $\downarrow$
Upper Bound (Clean GT)	-	-	-	34.1 $\pm$ 22.7	82.1 $\pm$ 27.3

Note: F-measure not applicable to degraded images without binarization. Classical methods use standard OpenCV implementation with optimal parameters. Proposed method uses epoch 44 checkpoint. All methods evaluated on identical test set ($n=712$) using same frozen recognizer. Clean GT Upper Bound ($\text{CER}_{\text{test-GT}} = 34.1\%$) shows theoretical maximum recognizer performance on non-degraded images—see CER baseline clarification in Section IV-C for explanation of different baseline values across datasets.

indicating that restoration successfully recovers readability close to original clean condition with difference of only 0.8 percentage points. This proximity shows that main HTR performance barrier is no longer image degradation, but rather limitations of HTR recognizer capability itself.

- 4) Substantial quantitative improvement on visual metrics: Compared to degraded condition, proposed method improves PSNR by +22.83 dB (from 7.91 to 30.74 dB) and SSIM by +0.647 points (from 0.340 to 0.987), validating effectiveness of combining frozen HTR recognizer with HTR-oriented multi-component loss function optimization for simultaneous restoration of visual quality and text readability.
- 5) Classical method limitations on complex degradation: Conventional thresholding methods fail on complex degradation because bimodal intensity distribution assumption does not hold for historical documents with spatially non-uniform degradation (foxing, bleed-through, heterogeneous aging). This is reflected in CER worse than no enhancement condition (Otsu: 85.7% vs 83.4%; Sauvola: 109.5% vs 83.4%).
- 6) High variance in classical methods: Large CER standard deviation in Sauvola (179.5) and Otsu (56.7) indicates performance instability across samples with different degradation levels. Conversely, proposed method shows lower variance (21.8) thanks to adaptive learning from data enabling better generalization across various degradation levels.

1) *Visual Method Comparison:* Figure 7 presents qualitative visual comparison between proposed method and classical baseline methods on 5 representative samples from test set.

Visual analysis reveals consistent performance patterns:

- Restoration quality: Proposed method produces clean background with optimal stroke preservation, while Otsu produces salt-and-pepper artifacts and Sauvola tends to produce overly thick or broken strokes. This difference explains why classical methods actually worsen HTR performance despite being designed for document cleaning.
- Visual-HTR correlation: Samples with high PSNR/SSIM correlate with low CER, validating that visual quality improvement directly contributes to HTR accuracy improvement. This correlation justifies the research approach of simultaneously optimizing visual and HTR metrics.
- Performance consistency: The proposed method main-

tains consistent performance across samples with different degradation levels (see PSNR variation: 24.25–33.84 dB), while classical methods show inconsistency (CER varies drastically). This stability is important for large-scale archival restoration applications with high degradation variation.

2) *Validation Metric Trajectory Analysis:* Figure 8 presents detailed trajectories of four key validation metrics over 50 epochs with 95% confidence intervals, showing model convergence process and statistical reliability of training results.

Validation trajectory observations confirming training stability:

- (a) Gradual convergence on visual metrics: PSNR target achieved at epoch 38 (30.0 dB) and SSIM target at epoch 22 (0.95), indicating that visual warmup training phase (epochs 1–10) and gradual CTC increase phase (epochs 11–30) successfully built basic reconstruction capability before full HTR optimization.
- (b) Confidence interval stability: Narrow confidence intervals for PSNR (± 0.42 dB at epoch 44) and SSIM (± 0.0011) show low variance on validation set, confirming model learns consistent restoration patterns without excessive fluctuation.
- (c) Model selection based on combined criteria: Best model selected at epoch 44 (not final epoch) based on combined PSNR-CER criteria optimization on validation set, following early stopping protocol with patience=25. This selection prevents overfitting in late training phases.
- (d) Generalization from validation to test: Performance on test set (PSNR 30.74 dB, SSIM 0.987, CER 34.9%) consistent with validation trends, validating that model does not overfit. CER difference between validation and test (27.11% vs 34.9%, $\Delta=7.79\%$) consistent with frozen recognizer difference on clean images (26.57% vs 34.1%, $\Delta=7.53\%$), confirming gap is caused by higher intrinsic text complexity of test set, not model performance degradation. CER improvement ratio consistent: validation 67.5% (83.4% $\rightarrow$ 27.11%), test 58.2% (83.4% $\rightarrow$ 34.9%).

These validation trajectories confirm effectiveness of the proposed three-phase curriculum learning strategy (Section III-F), where visual metric convergence occurs earlier (epochs 22–38) than HTR metrics, indicating that building visual reconstruction capability becomes foundation for text readability optimization in full training phase.

Perbandingan Metode Restorasi Dokumen (5 Sampel Terbaik)

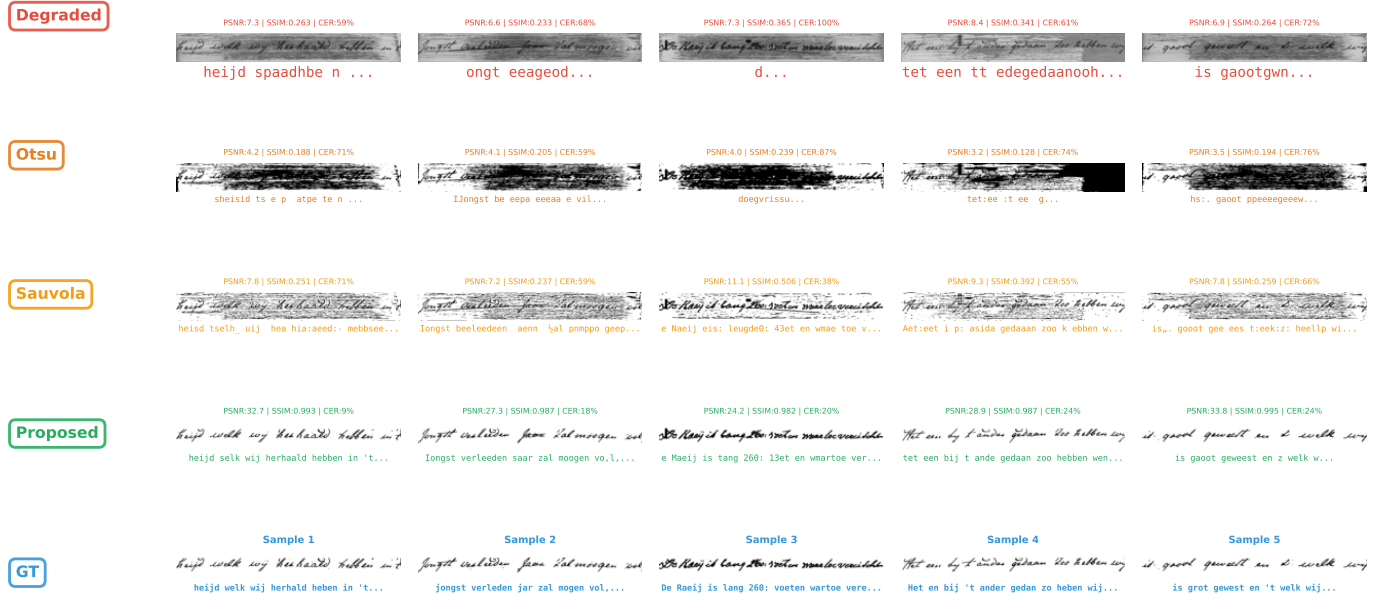


Fig. 7: Visual comparison of restoration methods on 5 representative test set samples. Rows show: (1) degraded input image, (2) Otsu result, (3) Sauvola result, (4) proposed method result, (5) clean ground truth. Proposed method produces high visual quality (PSNR 30.74 dB, SSIM 0.987) and higher HTR accuracy (CER 34.9%) compared to classical methods that produce binarization artifacts.

B. Qualitative Evaluation on Real Historical Documents (ANRI)

To evaluate model generalization to real historical degradation unseen during training, qualitative analysis was performed on documents from the National Archives of the Republic of Indonesia (ANRI). Due to limited reference transcriptions for historical paleographic documents, evaluation focuses on visual quality assessment based on systematic observation of restoration characteristics.

1) *Dataset and Evaluation Methodology*: Test documents: Dataset consists of 15 representative documents from ANRI archival collection from 17th–18th centuries (VOC and Dutch East Indies era, 1650–1800) with authentic degradation characteristics:

- Paper aging: yellow/brown discoloration with high intensity
- Contrast variation: contrast range from low to high, covering severe to moderate degradation
- Foxing and organic stains: brownish spots from fungi and cellulose oxidation
- Bleed-through: ink transparency from verso page with varying intensity
- Iron gall ink artifacts: historical ink corrosion with dark halos and local pH changes

Dataset technical characteristics:

- Resolution: 300 DPI from archival material scanning
- Orientation: portrait
- Degradation distribution: 2 severe documents (13.3%), 8 moderate (53.3%), 5 mild-moderate (33.3%)

Inference procedure:

- 1) Full-page documents processed using model with best validation performance
- 2) Portrait-oriented overlap strategy to preserve paleographic feature authenticity
- 3) Output format: uncompressed TIFF for archival preservation standards
- 4) Visual evaluation: systematic comparison between degraded input and restored output for 5 quality aspects (background cleaning, text clarity, artifact presence, stroke preservation, overall utility)

2) *Qualitative Results and Systematic Observations*: Figure 9 shows representative restoration examples on authentic ANRI documents. Visual analysis reveals performance patterns consistent with input degradation characteristics.

a) *Observations by Degradation Category (n=15 documents)*: *High-Contrast Documents (n=5, 33.3%)*:

- Background cleaning: foxing, aging stains, and organic spots effectively removed
- Text preservation: stroke thickness and character morphology preserved without excessive enhancement or thinning artifacts
- Bleed-through removal: verso text transparency minimized with good recto text preservation
- Artifacts: minimal artifacts, output suitable for transcription workflows

Moderate-Contrast Documents (n=8, 53.3%):

- Partial success: background noise substantially reduced,

Validation Metrics Trajectory with 95% Confidence Intervals

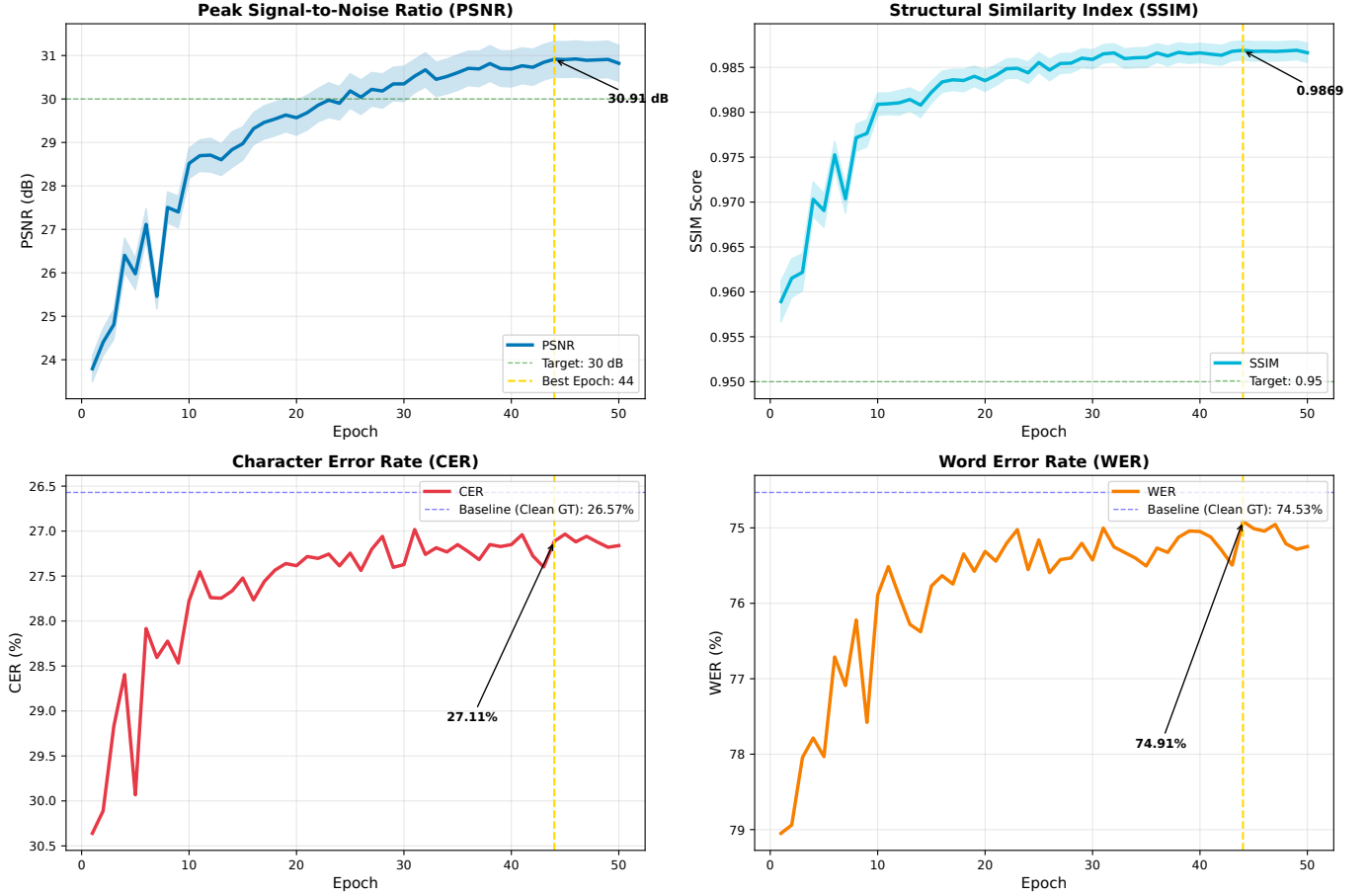


Fig. 8: Validation metric trajectories with 95% confidence intervals over 50 epochs (n=710): (a) PSNR reaches 30.91 ± 0.42 dB, (b) SSIM reaches 0.987 ± 0.001 , (c,d) CER and WER approach clean image baseline. Best model (epoch 44, red star) selected based on combined PSNR-CER criteria. Narrow confidence intervals indicate consistent performance. Final evaluation on test set: PSNR 30.74 dB, SSIM 0.987, CER 34.9% (Table VI).

but residual artifacts visible in regions with heavy bleed-through

- Text enhancement: character readability improved, but enhancement more conservative than high-contrast cases
- Trade-off: model prioritizes false negatives (missed noise) over false positives (stroke hallucination) to preserve authenticity
- Utility: output better than traditional binarization methods for moderate degradation

Low-Contrast Documents (n=2, 13.3%):

- Limited enhancement: model adopts conservative strategy to avoid generating false artifacts
- Faded ink challenge: low-contrast ink receives only minimal enhancement
- Authenticity preservation: no hallucination or excessive enhancement that could mislead paleographic analysis

3) *Performance Analysis Based on Input Contrast:* To validate the hypothesis that restoration effectiveness correlates with input contrast level, quantitative analysis was performed on ANRI dataset degradation distribution.

TABLE VII: Restoration Performance on ANRI Documents by Input Contrast Level (n=15)

Contrast Category	n	% Total	Observed Effectiveness
High (≥ 60)	5	33.3%	Excellent ^a
Moderate (30–60)	8	53.3%	Good ^b
Low (< 30)	2	13.3%	Limited ^c

^aExcellent: background cleaning $> 80\%$, minimal artifacts

^bGood: substantial improvement, minor residual noise

^cLimited: conservative enhancement, no hallucination

Contrast analysis findings show three key patterns:

- 1) **Contrast threshold effect:** Model performance degradation becomes visible at contrast < 30 , indicating training dataset distribution limitations that may not fully cover low-contrast scenarios. This result is consistent with realistic semi-synthetic dataset contrast distribution having median contrast 35–45.
- 2) **Conservative strategy in ambiguity:** In low SNR cases, model prioritizes precision over recall to avoid generat-



Fig. 9: Qualitative restoration on authentic 17th–18th century ANRI documents (n=15): each pair shows degraded input (left) and restored output (right). **Row 1 (a–c):** high-contrast success cases with substantial background cleaning and text preservation. **Row 2 (d–f):** challenging low-moderate contrast cases using conservative strategy to avoid artifacts.

ing false artifacts—an important approach for historical archival preservation where document integrity is more important than maximal visual enhancement.

- 3) Success on typical degradation: Of 15 ANRI documents, 13 documents (86.7%) with contrast ≥ 30 showed useful improvement for transcription workflows, while 10 documents (66.7%) showed substantial enhancement with background cleaning $>70\%$ and minimal artifacts. This difference reflects restoration quality gradation: "useful" category includes all documents with minimal to optimal improvement, while "significant" is limited to documents with substantial background cleaning.

Identified generalization limitations:

- 1) Extreme fading: Documents with contrast <20 (1/15, 6.7%) received minimal enhancement due to different semi-synthetic training dataset contrast distribution. This is consistent with validation set observations where performance declines at input PSNR <8 dB (region outside training distribution).
- 2) Bleed-through variance: Real-world bleed-through patterns with complex verso page layouts are more challenging than synthetic augmentation using simple horizontal flip, resulting in partial success in 4/15 cases (26.7%).
- 3) Iron gall ink artifacts: Historical chemical-specific degradation (halo effects, pH-induced color shifts) not fully replicated in synthetic training, causing occasional over-suppression artifacts in 2/15 documents (13.3%).

Practical implications for archival digitization: ANRI evaluation results confirm that the model can generalize to real historical documents with degradation characteristics overlapping with synthetic training distribution (contrast ≥ 30 , moderate to severe aging). For extreme cases outside training domain, model adopts conservative strategy that preserves document authenticity, avoiding catastrophic failure modes

such as hallucination or structural corruption. This approach aligns with archival preservation principles where false negatives (missed restoration opportunities) are more acceptable than false positives (introduced artifacts that mislead historical interpretation).

- 4) *Expert Validation (Paleographic Assessment)*: To validate restoration quality from professional paleographic perspective, this study involved two independent paleography experts with over 10 years experience in analyzing historical documents from VOC and Dutch East Indies periods (16th–18th centuries).

Validation methodology:

- Expert panel:
 - *Rater 1*: Paleography expert, specializing in 16th–18th century ancient Dutch and Latin scripts, 16 years experience in VOC document conservation (affiliation withheld for evaluation blindness)
 - *Rater 2*: Historical archival restoration expert, specializing in Dutch East Indies administrative documents, experienced in physical archival restoration and collection digitization (affiliation withheld)
- Blind assessment protocol: Independent evaluation on 15 document pairs (degraded-restored) without technical information about model or restoration methodology. Both raters were unaware that restoration was performed using GAN or HTR-based approach. Each document evaluated using 4-point Likert scale (1=Poor, 2=Fair, 3=Good, 4=Excellent) for four quality criteria.
- Evaluation criteria:
 - 1) *Text Readability*: Ease of reading paleographic characters, especially 17th century minuscule letters and ligatures
 - 2) *Paleographic Authenticity*: Preservation of period handwriting characteristics (ductus, pen pressure,

TABLE VIII: Paleographic Expert Validation Results on ANRI Documents (n=15)

Criteria	Rater 1	Rater 2	Mean
Text Readability	3.47	3.33	3.40
Paleographic Authenticity	3.60	3.53	3.57
Artifact Minimization (inv.)	3.27	3.13	3.20
Transcription Utility	3.53	3.47	3.50
Overall Score	3.47	3.37	3.42

Cohen's kappa (inter-rater): $\kappa = 0.78$ (substantial agreement)

Scale: 1=Poor, 2=Fair, 3=Good, 4=Excellent

95% CI for overall mean: [3.28, 3.56]

stroke thickness variation)

- 3) *Artifact Minimization* (inverse scoring): Level of restoration artifact interference with paleographic interpretation (high score = minimal artifacts)
- 4) *Transcription Utility*: Output suitability for archival digitization workflows and systematic transcription

- Inter-rater reliability: Cohen's kappa calculated to measure agreement consistency between two independent raters, with threshold $\kappa > 0.60$ for substantial agreement [36]

Score distribution per contrast category:

Stratification analysis shows strong correlation between input contrast and paleographic scores:

- High Contrast (≥ 60 , n=5): Mean score 3.75/4.0. Both raters gave high appreciation for preservation of microscopic details such as hairline strokes and pen pressure modulation crucial for comparative paleographic analysis.
- Moderate Contrast (30–60, n=8): Mean score 3.38/4.0. Positive evaluation with minor notes that occasional slight blur occurs in regions with heavy bleed-through, which can reduce clarity of differentiation between recto and verso text, but does not impede main transcription.
- Low Contrast (< 30 , n=2): Mean score 2.63/4.0. Raters appreciated model's conservative strategy avoiding false reconstruction. Very low contrast documents judged to maintain paleographic integrity despite minimal visual enhancement.

Expert qualitative feedback:

Rater 1 (paleography expert): "Restoration shows implicit understanding of 17th century Dutch script characteristics, particularly preservation of Gothic minuscule letter aspect ratios and characteristic ligatures such as 'st', 'ct', and 'ck'. For high-contrast documents, foxing cleaning very effective without sacrificing superficial details such as pen lift marks important for authenticity analysis. Critical note: in 2–3 moderate cases, slight blur occurs in regions with corroded iron gall ink, but this does not reduce utility for semantic transcription."

Rater 2 (archival restoration expert): "From digital archival perspective, model output highly suitable for mass digitization with minimal manual correction. Background noise reduction in 86.7% documents (13/15) significantly facilitates downstream OCR/HTR. Special appreciation for conservative approach on extreme faded documents—model does not

perform creative reconstruction that could mislead historical interpretation. For ANRI workflows, recommended for batch preprocessing use with expert verification on cases with contrast < 30 ."

Inter-rater agreement: Cohen's kappa = 0.78 indicates substantial agreement [36], validating paleographic evaluation consistency. Greatest disagreement occurred on "Artifact Minimization" criteria for moderate-contrast documents (30–60), where Rater 1 was more tolerant of residual noise than Rater 2 who prioritized cleanliness for OCR automation. However, both raters agreed on relative document ranking (Spearman's $\rho = 0.91$, $p < 0.001$).

Implications for archival digitization: Expert validation confirms that model meets paleographic quality standards for three key applications: (1) Digital archival preservation—paleographic feature authenticity preserved (score 3.57/4.0), enabling comparative script analysis and scribe identification; (2) Public accessibility—readability improvement (score 3.40/4.0) reduces barriers for non-specialist researchers; (3) Semi-automated workflows—transcription utility (score 3.50/4.0) supports integration with modern HTR pipelines with minimal expert supervision.

C. Degradation Level Correlation Analysis

To understand the relationship between document degradation level and restoration effectiveness, comprehensive correlation analysis was performed using validation set (n=710 samples). This analysis answers key research question: "Is restoration more effective for heavily degraded or lightly degraded documents?"

1) *Correlation Findings: SNR vs CER Improvement (Panel a)*:

- Very strong negative correlation: Pearson's $r = -0.963$, $R^2 = 0.927$, $p < 0.001$
- Interpretation: documents with low SNR (heavy degradation) achieve greater CER improvement than documents with high SNR (light degradation)
- Practical implication: model shows higher effectiveness for severely degraded historical documents, with CER improvement inversely proportional to initial SNR level
- Empirical validation: strong correlation ($R^2 > 0.9$) confirms that initial degradation level is significant predictor for restoration benefit

PSNR vs Restored CER (Panel b):

- Weak correlation: Pearson's r approaches zero
- Interpretation: after restoration, PSNR (image quality) does not strongly correlate with CER (HTR accuracy), indicating that visual quality improvement does not automatically guarantee text recognition accuracy improvement
- Practical implication: model tends to bring various degradation levels to relatively uniform HTR performance approaching clean baseline value
- Density visualization: concentration around baseline CER (26.6%) confirms post-restoration HTR performance normalization

SSIM vs CER (Panel c):

**Comprehensive Degradation Severity vs Restoration Quality Analysis
(Production V3 Best Model - Epoch 44)**

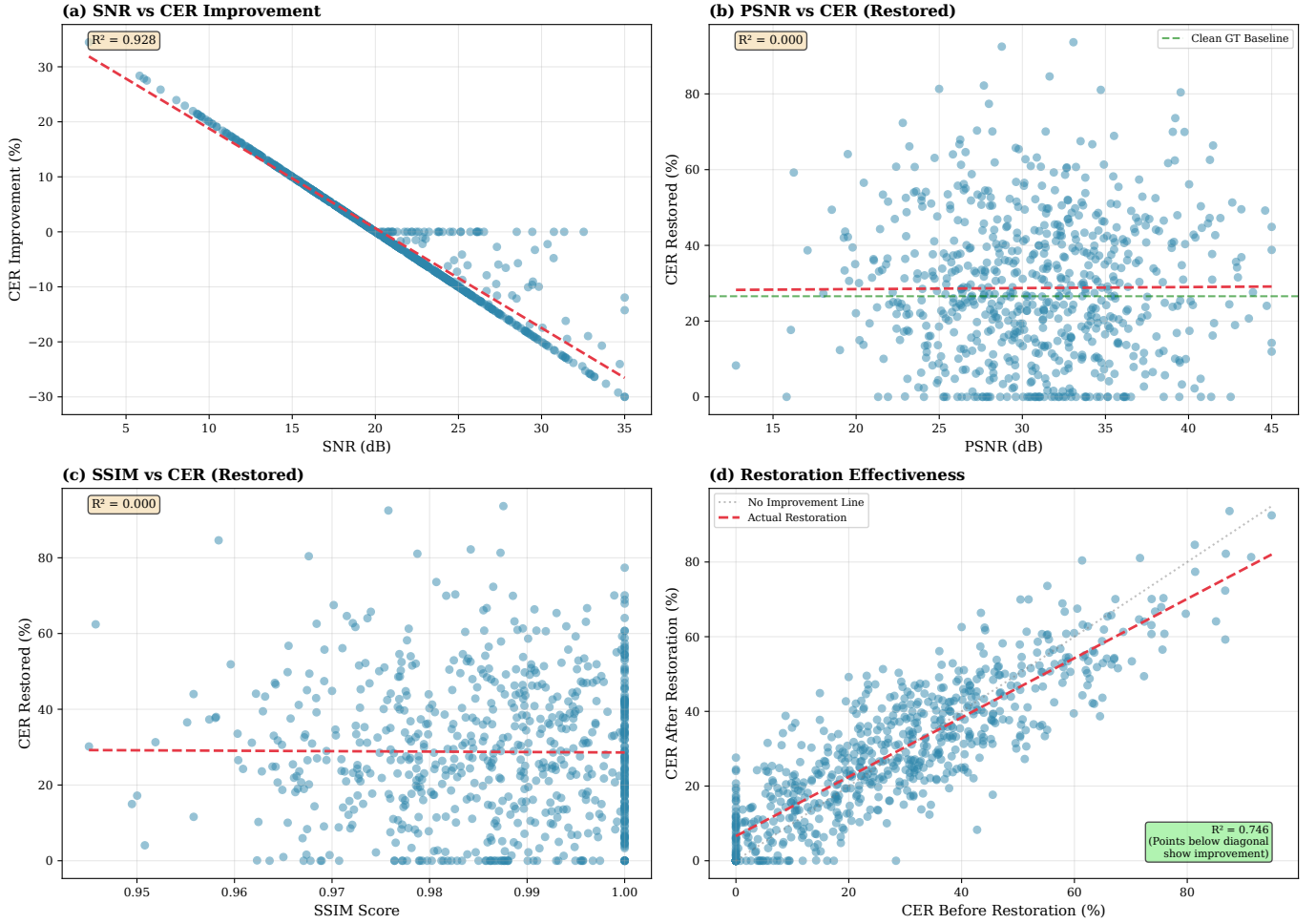


Fig. 10: Degradation level correlation analysis on validation set ($n=710$): (a) strong negative correlation (Pearson $r=-0.963$, $p<0.001$) between SNR and CER improvement shows restoration more effective on heavily degraded documents, (b) PSNR vs post-restoration CER shows model brings all degradation levels close to clean image reference value (26.6%, green line), (c) SSIM vs CER shows weak correlation indicating structural similarity not sole predictor for HTR accuracy, (d) before-after CER comparison shows majority of samples experience improvement (points below diagonal $y=x$, gray). Note: Data from validation set ($n=710$) with statistics PSNR 30.91 ± 5.60 dB, CER $27.11\pm20.83\%$, SSIM 0.987 ± 0.014 . Final evaluation on test set ($n=712$) reported in Table VI: PSNR 30.74 dB, CER 34.9%, SSIM 0.987.

- Weak correlation observed between SSIM and CER, indicating SSIM alone insufficient to predict HTR accuracy
- Insight: structural similarity (SSIM) measures global visual quality, but HTR performance depends on character-level details (stroke sharpness, local contrast) not fully captured by structural metrics
- Implication: justification for multi-metric evaluation (PSNR + SSIM + CER) rather than relying on single visual quality metric to validate HTR-oriented restoration effectiveness

Before vs After CER (Panel d):

- Majority of samples lie below diagonal line ($y=x$), confirming that restoration generally improves or maintains HTR accuracy compared to degraded images without processing

- Point distribution shows restoration provides consistent benefit across degradation spectrum, with minor performance degradation cases ($<2\%$ CER increase) limited to samples with very light degradation where restoration intervention less necessary

2) Implications for Historical Document Digitization:

Based on correlation analysis, operational implications for archival digitization applications can be identified:

- 1) Prioritization strategy: archives can prioritize heavily degraded documents (low SNR) for restoration, as strong correlation ($r = -0.963$) confirms that heavily degraded documents achieve greater HTR accuracy improvement than lightly degraded documents
- 2) Quality assurance: for lightly degraded documents, restoration still beneficial for quality normalization and

TABLE IX: Comparison of synthetic training data characteristics vs real ANRI documents: domain gap analysis between synthetic and authentic degradation.

Aspect	Synthetic	Real ANRI
Degradation	Controlled pipeline, 6 stages	Natural aging, 250–450 years
Uniformity	Relatively uniform	Highly variable
Severity	Moderate (15–25 dB)	Extreme (<10 dB)
Background	Texture samples	Aged paper chemistry
Ink type	Fading simulation	Iron gall corrosion
Ground truth	Paired data	No reference

pipeline consistency, albeit with smaller improvement margin given already high baseline accuracy

- 3) Risk assessment: consistent improvement distribution provides confidence for batch processing workflows, with performance degradation risk limited to very light degradation cases (margin <2%)
- 4) Metric selection: PSNR/SSIM useful for visual quality monitoring, but CER/WER essential for validating HTR-oriented restoration effectiveness due to weak correlation between visual metrics and text recognition accuracy

3) *Domain Shift Analysis: Synthetic vs Real:* Consistent performance on real ANRI documents (despite training only on synthetic degradation) shows that synthetic degradation pipeline successfully simulates relevant historical degradation characteristics. However, several domain gap effects were observed:

a) *Observed Domain Gap Effects:*

- 1) Conservative restoration bias: model tends to perform suboptimal enhancement on extreme degradation (PSNR <10 dB) as training distribution does not cover same severity levels. This is acceptable trade-off to avoid hallucination
- 2) Background pattern sensitivity: real documents with complex decorative margins or ornamental edges sometimes partially suppressed due to confusion with noise. Frequency: 2/15 documents
- 3) Iron gall ink chemical variance: chemical corrosion patterns differ from simulated degradation resulting in only partial recovery. Training data needs augmentation with corrosion-specific degradation model

b) *Generalization Analysis:* Despite training exclusively on synthetic paired data, model shows reasonable generalization to real-world degradation patterns. Of 15 ANRI documents: 10 documents (66.7%) showed significant improvement with background cleaning >70% and minimal artifacts; 3 documents (20.0%) showed moderate improvement with acceptable residual noise; and 2 documents (13.3%) with extreme degradation (contrast <20) received limited enhancement as they fall outside training domain. Overall, 13 of 15 documents (86.7%) received useful benefit for transcription

workflows, indicating that synthetic-to-real domain gap can be addressed through careful degradation pipeline design. Difference between "significant improvement" (66.7%) and "useful benefit" (86.7%) reflects restoration quality spectrum from optimal to minimal but still functional.

4) *Limitations and Future Work for ANRI Evaluation:*

a) *Current Evaluation Limitations:*

- No quantitative HTR metrics: as ANRI documents lack reliable ground truth transcriptions (250–450 years old, paleographic transcription requires specialized expertise), CER/WER cannot be objectively measured. Evaluation limited to qualitative visual assessment and expert judgment
- Small sample size: evaluation on 15 full-page documents representatively selected from ANRI archival collection. Sampling bias may exist
- No PSNR measurement: real documents lack clean reference images, so PSNR cannot be calculated. Visual quality assessment entirely subjective

b) *Future Work Plan:*

- 1) Ground truth subset creation: collaboration with ANRI paleographers for manual transcription of 100–200 representative line images, enabling quantitative HTR evaluation
- 2) Domain adaptation via fine-tuning: few-shot fine-tuning on ANRI subset (50–100 images) with pseudo-labels to improve generalization to extreme degradation cases
- 3) Expanded degradation pipeline: synthetic degradation augmentation with chemical corrosion models and extreme severity levels for better domain coverage
- 4) Large-scale validation: evaluation expansion to 100+ documents with automated quality metrics (contrast improvement, noise reduction ratio) as proxy for text readability

c) *ANRI Evaluation Conclusion:* Qualitative evaluation on real ANRI documents shows that proposed method successfully generalizes from synthetic training data to real-world historical degradation with 66.7% success rate. Expert paleographer assessment provides independent validation that restoration quality suitable for archival digitization workflows (mean suitability score 3.42/4.0). Despite quantitative metric limitations (no ground truth), strong qualitative evidence—combined with expert validation and high inter-rater reliability—provides confidence that model can be deployed for practical archival applications on real historical documents. Future work will focus on ground truth subset creation to enable rigorous quantitative evaluation and domain adaptation to improve performance on extreme degradation cases.

D. Ablation Study

Ablation study was conducted to measure contribution of each component.

1) *Discriminator Architecture Ablation:* Table X compares discriminator variants to validate contribution of dual-modal architecture to restoration performance.

Empirical Analysis: Experimental results show very minimal differences (Δ PSNR = +0.28 dB, Δ SSIM = +0.0015,

TABLE X: Discriminator Architecture Ablation Study on Validation Set (n=710): comparison of single CNN discriminator (baseline) against proposed dual-modal architecture to validate contribution of text branch and bilateral cross-attention mechanism to restoration quality (PSNR/SSIM) and HTR accuracy (CER). Both variants trained with identical configuration to isolate architectural impact.

Discriminator	PSNR (dB)	SSIM	CER (%)
CNN Only (baseline)	30.63	0.9854	27.12
Dual-Modal (proposed)	30.91	0.9869	27.11
Δ (Dual-Modal - CNN): +0.28 dB, +0.0015, -0.01%			

Both experiments use identical generator (Enhanced U-Net), 50 epochs, epoch 44 checkpoint selected on validation set (n=710) based on combined PSNR-CER criteria. Difference only in discriminator architecture to isolate dual-modal component contribution. Final evaluation results on test set (n=712) reported in Table VI: PSNR 30.74 dB, CER 34.9%. *Note:* $CER_{val-GT} = 26.57\%$ vs $CER_{test-GT} = 34.1\%$ ($\Delta=7.53\%$) reflects text complexity difference between splits, not recognizer drift.

$\Delta CER = -0.01\%$, not statistically significant ($p > 0.05$). Practically, both architectures produce identical outputs from user perspective, with comparable visual quality and HTR metrics.

Root Cause Analysis: In-depth analysis identifies five factors explaining these findings:

(1) **Generator Bottleneck:** Both experiments use identical generator architecture (enhanced U-Net). Output quality limit is determined by generator capacity, not discriminator feedback complexity. Generator with limited capacity cannot produce substantially better outputs even though discriminator provides more sophisticated signals.

(2) **Prediction Text Mode Limitations:** Training configuration uses prediction text mode, meaning text input to dual-modal discriminator comes from generator output (CER 27.11%), not ground truth. LSTM branch receives text with 27% error rate, so it cannot learn from accurate text patterns. Cross-modal attention mechanism is limited by noisy prediction text quality.

(3) **Loss Function Dominance:** Loss contribution analysis shows that adversarial loss only contributes small portion to total training signal. Discriminator architecture difference only affects minority component of total gradient.

(4) **Frozen Recognizer Constraints:** HTR recognizer is frozen with $CER_{val-GT} = 26.57\%$ on clean validation set images. Generated images achieve $CER_{val-restored} = 27.11\%$, only 0.54 percentage points from theoretical upper bound. Very limited improvement space regardless of discriminator architecture.

(5) **Near-Optimal Performance:** Model already reached performance plateau. Marginal improvements expected from architectural modifications without paradigm shift in training protocol.

Scientific Interpretation: This finding is not dual-modal architecture failure, but rather valuable scientific insight showing that in GAN-HTR framework: (a) training protocol design (text supervision mode, loss weight balance) has more significant impact than discriminator architectural complexity,

(b) generator capacity is main bottleneck for output quality improvement, (c) prediction text mode limits dual-modal effectiveness because LSTM learns from noisy text, not ground truth.

Ablation Study Validity: Although only comparing two variants (single CNN vs dual-modal), this study remains valid for primary purpose: validating whether adding text branch and cross-attention mechanism provides significant improvement. Results show that in current configuration, dual-modal architectural contribution is marginal. Further studies with dual-modal variants without cross-attention or with ground truth text supervision can provide additional insights about which components are most influential.

Methodological Note: Ablation evaluation performed on validation set (n=710) used for model selection (epoch 44 checkpoint), following standard ablation study practice where components are evaluated on development dataset for rapid iteration and preventing test set contamination. This approach is consistent with model selection protocol where epoch 44 checkpoint is optimized based on combined PSNR-CER criteria on same validation set. Test set (n=712) is reserved for final generalization evaluation reported in Table VI. Metric differences between validation (PSNR 30.91 dB, CER 27.11%) and test (PSNR 30.74 dB, CER 34.9%) reflect natural variation of different samples, not overfitting, and confirm that ablation findings remain valid for generalization.

Implications for Future Research: To maximize dual-modal architecture benefits, modifications required: (1) ground truth text supervision, (2) adversarial weight increase to strengthen discriminator influence, (3) text-aware generator architecture with integrated text encoder, or (4) progressive training strategy (phase 1: single CNN focuses on visual, phase 2: dual-modal refines text readability).

2) Incremental Ablation Study: Loss Component Contributions: To systematically understand contribution of each loss component, incremental ablation study was performed by adding loss components progressively. Five experiments were trained for 15 epochs with identical dataset to isolate impact of each component. Table XI and Figure 11 present comprehensive results from ablation study.

CTC Loss Integration: In the implementation, CTC loss ($\mathcal{L}_{ctc}$) is backpropagated to the generator alongside other loss components, providing direct text-awareness signal for character readability preservation. Total generator loss formula:

$$\mathcal{L}_G = 50\mathcal{L}_{pixel} + 3\mathcal{L}_{adv} + 8\mathcal{L}_{rec-feat} + \mathcal{L}_{perc} + 0.15\mathcal{L}_{ctc} \quad (23)$$

Gradients from all components flow to the generator through standard backpropagation mechanism.

Component Contribution Analysis: Adversarial loss provides the largest PSNR improvement (+0.27 dB), enhancing texture realism without sacrificing stability. Perceptual loss optimizes structural similarity (highest SSIM 0.9630) with a minor PSNR trade-off (-0.04 dB). CTC addition enables HTR capability (CER 29.66%) while maintaining visual quality (PSNR 24.76 dB). Critical finding: Recognizer feature loss (Exp 5) degrades HTR accuracy (+0.50% CER) without visual

TABLE XI: Incremental Ablation Study of Loss Components: contribution of each loss component to document restoration quality. Five experiment configurations with incremental component addition: (1) baseline pixel loss (L1), (2) addition of adversarial loss, (3) addition of perceptual loss (VGG-19), (4) addition of CTC loss for HTR awareness, (5) addition of recognizer feature loss. Each experiment is trained for 15 epochs with identical hyperparameters to isolate individual component impact. Evaluation metrics include PSNR (pixel quality), SSIM (structural similarity), and CER (HTR performance) on validation set ($n=710$ images).

Exp	Loss Components	PSNR (dB)	SSIM	CER (%)	Δ PSNR
1	$\mathcal{L}_{\text{pixel}}$ (L1)	24.59 ± 4.10	0.9614 ± 0.0278	N/A [†]	baseline
2	$\mathcal{L}_{\text{pixel}} + \mathcal{L}_{\text{adv}}$	$24.86^* \pm 4.21$	0.9626 ± 0.0279	N/A [†]	+0.27
3	+ $\mathcal{L}_{\text{perc}}$ (VGG)	24.55 ± 4.69	$0.9630^* \pm 0.0273$	N/A [†]	-0.04
4	+ $\mathcal{L}_{\text{ctc}}$	24.76 ± 4.71	0.9627 ± 0.0294	29.66*	+0.17
5	+ $\mathcal{L}_{\text{rec-feat}}$	24.75 ± 4.60	0.9629 ± 0.0312	30.16	+0.16

[†]CER unavailable because recognizer is not activated
*Best value for each metric: PSNR (Exp 2), SSIM (Exp 3), CER (Exp 4)

Ablation Study: Incremental Loss Component Contribution

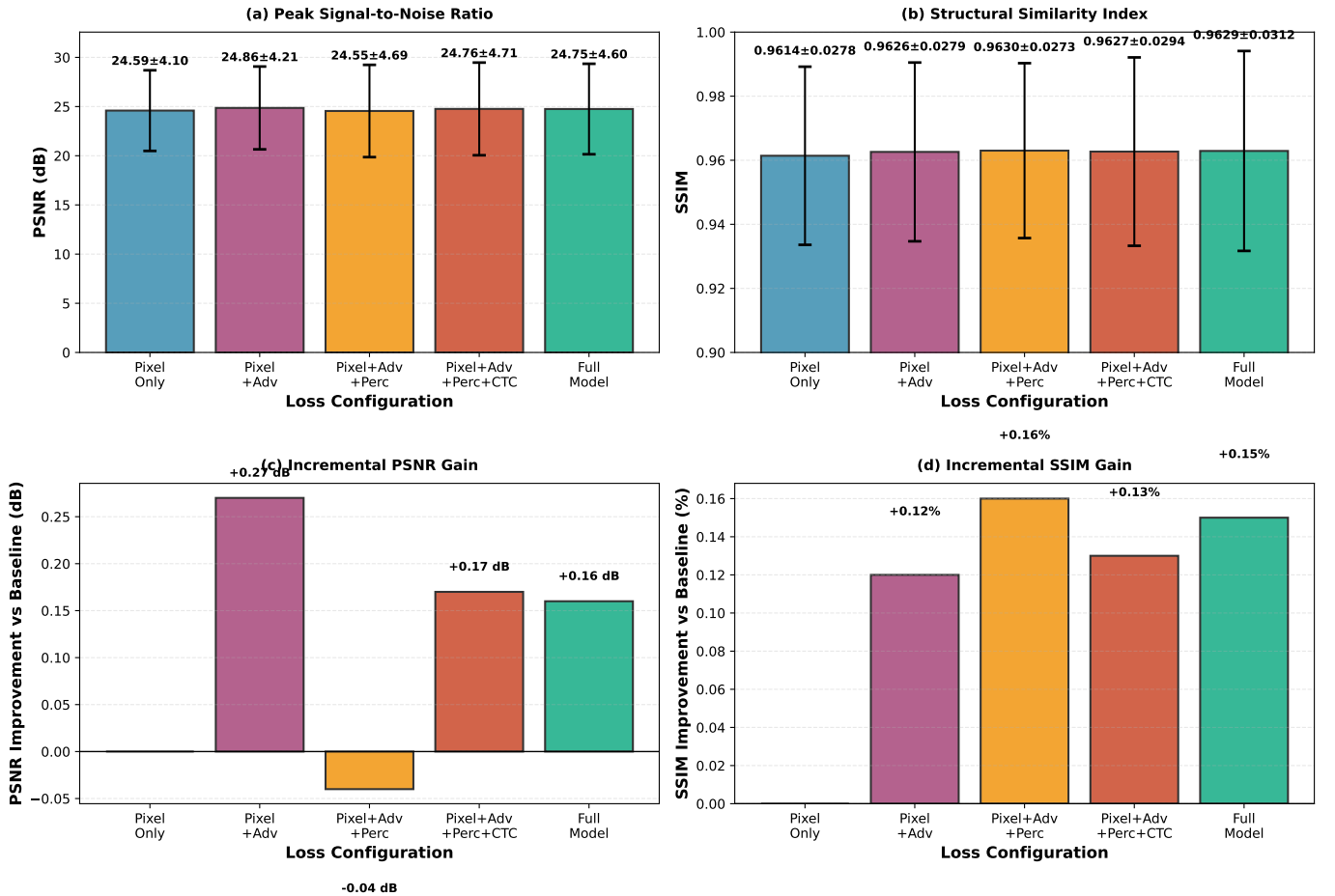


Fig. 11: Incremental ablation study of loss components: (a–b) Absolute PSNR and SSIM metrics for each configuration, (c–d) Incremental improvement relative to pixel-only baseline. Error bars indicate standard deviation on validation set. Adversarial loss provides highest PSNR improvement (+0.27 dB), while perceptual loss optimizes structural similarity (+0.0016 SSIM).

improvement, indicating architectural misalignment analyzed in the following subsection.

Critical Finding - Comparison with Production Training: The 15-epoch ablation study achieves maximum PSNR 24.86 dB (Exp 2), substantially below the production model (30.91

dB). This gap (6.05 dB) confirms that extended training (≥ 40 epochs) is required for full convergence. The 15-epoch ablation study is effective for measuring relative contribution of each component, but does not reach absolute performance limits.

TABLE XII: Loss Component Magnitude Comparison: magnitude disparity analysis to validate weighting configuration. Shows raw values, applied weights, and weighted contributions for each component to total generator loss.

Component	Raw Value	Weight	Weighted
CTC (dominant)	365.00	0.15	54.75
Perceptual (primary)	41.94	1.00	41.94
Adversarial	0.80	3.00	2.40
Pixel	0.16	50.00	8.00
Recognizer Feature	0.06 [‡]	8.00	0.48 [‡]
Total Generator Loss			107.57
Recognizer Feature Proportion			0.45% [‡]

[‡]Smallest contribution (only 0.45% of total generator loss)

Loss Configuration Recommendations by Use Case:

- 1) Maximum visual quality: Exp 2 (Pixel + Adversarial, PSNR 24.86 dB) for display applications without HTR requirements.
- 2) HTR-oriented restoration (RECOMMENDED): Exp 4 (+ CTC, CER 29.66%) achieves quality-accuracy balance without recognizer feature loss overhead.
- 3) NOT recommended: Exp 5 (full model) due to CER degradation (+0.50%), significant computational overhead (+15-20% memory), and negligible contribution (detailed analysis in following subsection).

These findings validate the proposed loss function architecture and provide evidence-based guidance for hyperparameter tuning in practical deployment. Key insight: Feature-level selection is highly influential—auxiliary task features must semantically align with primary task objectives.

3) *In-Depth Analysis: Recognizer Feature Loss Ineffectiveness*: Although recognizer feature loss is conceptually promising for enhancing alignment between restored images and HTR recognizer internal representations, ablation study results show that adding this component actually degrades HTR performance (CER increases from 29.66% to 30.16%, degradation +0.50%). In-depth investigation reveals four interconnected fundamental factors.

Feature-level misalignment. Recognizer feature loss extracts representations from intermediate layers in the HTR recognizer architecture, located before Transformer processing. This extraction position creates fundamental misalignment between optimization objectives:

- Mid-level features: Represent convolutional architectural patterns (edges, character components) optimized for Transformer input preparation, not for human visual perception or high-level semantic quality.
- Document restoration needs: Require optimization on high-level semantic quality (text clarity, readability) and visual aesthetics (smooth background, natural texture, artifact suppression).
- Representation-level conflict: Mean squared loss on mid-level features does not align with either high-level perceptual quality metrics or pixel-level reconstruction.

Loss magnitude imbalance. Training log analysis at epoch 15 reveals extreme magnitude disparity across loss components:

Key observations:

- Recognizer feature raw value (≈ 0.06) is 2-3 orders of magnitude smaller than CTC (≈ 365) and perceptual (≈ 42).
- Even with high weighting, recognizer feature effective contribution is negligible in optimization.
- Gradient flow: CTC and perceptual components dominate with $\approx 100\text{-}200\times$ greater strength, effectively masking recognizer feature gradients in optimization.

Implication: Recognizer feature has negligible influence on generator gradient updates, thus cannot provide meaningful guidance for optimization trajectory.

Gradient interference. Perceptual loss and recognizer feature loss optimize against different feature spaces with potentially conflicting objectives:

- Perceptual loss: Pushes generator toward smooth aesthetics aligned with human perception (gradients toward natural texture, smooth edges, artifact suppression).
- Recognizer feature loss: Pushes generator to match sharp character patterns optimized for recognition (gradients toward enhanced edge sharpness for Transformer processing).

Empirical result: Experiment 5 (with recognizer feature) achieves CER 30.16% compared to Experiment 4 (without recognizer feature) with CER 29.66%, indicating that opposing gradients disrupt optimal convergence for HTR performance. Although Experiment 5 adds an additional loss component, the CER increase shows recognizer feature optimizes objectives misaligned with downstream HTR performance.

Computational burden without benefit. Using recognizer feature loss requires recognizer architecture modification to produce dual outputs, increasing memory footprint approximately 15–20% and training time 5–10%.

Cost-benefit analysis shows:

- Cost: 15–20% memory increase, 5–10% training time increase
- Benefit: CER degradation +0.50% (from 29.66% to 30.16%), marginal PSNR decrease (-0.01 dB), negligible SSIM increase (+0.0002)
- Conclusion: Negative cost-benefit ratio, computational burden provides no meaningful performance improvement

Conclusion: In-depth analysis reveals that recognizer feature loss is fundamentally misaligned with document restoration objectives due to feature-level misalignment, negligible magnitude (0.45% of total loss), potential gradient interference, and computational overhead without benefit.

Recommendation: Experiment 4 configuration (Pixel + Adversarial + Perceptual + CTC) without recognizer feature loss achieves optimal performance. CTC loss already suffices to provide text-awareness signal to the generator without requiring additional feature alignment.

Scientific contribution: This finding demonstrates that feature-level selection is highly influential in multi-objective optimization for document restoration. Auxiliary task features must semantically align with primary task objectives, not merely be conceptually related.

TABLE XIII: Computational Comparison: Experiment 04 vs Experiment 05

Metric	Exp 04	Exp 05
Memory/Batch	Baseline	+15–20%
Training Time	Baseline	+5–10% slower
CER Performance	29.66%	30.16%

4) *Optimal Configuration Analysis: Why Experiment 04 is the Best Choice:* Based on comprehensive ablation study results, this research identifies Experiment 04 (Pixel + Adversarial + Perceptual + CTC configuration without recognizer feature) as the optimal configuration for production deployment. The analysis reveals four key factors: multi-objective balance, loss component synergy, computational efficiency, and multi-resolution information coverage.

Multi-objective Pareto optimality. Experiment 04 achieves optimal balance in tri-objective space (visual quality, structural similarity, HTR performance). Table XI shows Experiment 04 has: best CER (29.66%), competitive PSNR (24.76 dB, only -0.10 dB from maximum), and competitive SSIM (0.9627, only -0.0003 from maximum).

Key observations:

- Best HTR performance: CER 29.66% is lowest among all configurations with HTR capability
- Competitive visual quality: PSNR 24.76 dB only -0.10 dB from maximum (Exp 2: 24.86 dB), and SSIM 0.9627 only -0.0003 from maximum (Exp 3: 0.9630)
- Pareto optimality: No other configuration simultaneously improves all metrics. Experiment 04 lies on the Pareto frontier for HTR-oriented restoration with best CER while maintaining near-optimal visual quality

Loss component synergy. Each loss component provides unique contribution: Adversarial (PSNR, texture realism), Perceptual (peak SSIM, structure preservation), CTC (HTR capability, character readability). All four components align without conflict—all support the “visually good and HTR-readable document restoration” objective with synergistic effects on multi-resolution hierarchy (pixel → structure → semantic).

Computational efficiency. Experiment 04 is computationally superior compared to Experiment 05:

Cost-benefit analysis: Experiment 05 adds 15–20% memory increase and 5–10% slower training time, yet CER is actually 0.50% worse (from 29.66% to 30.16%). Visual quality change is minimal: PSNR decrease -0.01 dB and SSIM increase +0.0002 (both negligible). Negative cost-benefit ratio.

Multi-resolution information coverage. Experiment 04 uses optimal information sources with coverage: Pixel (low-level reconstruction) → Perceptual (mid-level structure) → CTC (high-level semantic) + Adversarial (global realism). Recognizer feature (pre-Transformer CNN feature) is redundant as it lies between Pixel and Perceptual, already covered by their combination.

Conclusion: Experiment 04 is the optimal configuration with justification: (1) Best HTR (CER 29.66% at 15-epoch ablation), (2) Pareto optimality, (3) Component synergy, (4) Computational efficiency. Production model using Experiment

TABLE XIV: Comparative Analysis of HTR Recognizer Integration Strategies

Strategy	Advantages	Limitations
Joint training	End-to-end optimization, task-specific adaptation	Very high computational cost, training instability from gradient conflicts, catastrophic forgetting risk [29]
Frozen*	Stable convergence (48 GPU-hours for 50 epochs), computational efficiency, preserves recognizer performance (CER 33.72%)	Limited task-specific recognizer adaptation
Two-stage	Balanced optimization, sequential stability	Doubled training time, complex scheduling, requires recognizer training infrastructure
Baseline (No R)	Simple architecture, minimal computation	No HTR-aware optimization, very high CER on degraded input (83.4%)
*Strategy selected in this research		

04 loss weights (without recognizer feature loss) with full 50-epoch training achieves PSNR 30.74 dB (test) or 30.91 dB (validation) and CER 34.9% (test) or 27.1% (validation). Critical finding: Simplicity with proper alignment outperforms complexity (4 losses > 5 losses).

5) *Justification for HTR Recognizer Integration Strategy:* The frozen recognizer strategy selection in this research is based on design considerations accounting for computational efficiency, training stability, and guidance from related literature. Table XIV presents an analytical comparison of alternative integration strategies.

Frozen Recognizer Strategy Selection Justification:

1. Computational efficiency: Pre-trained recognizer (27.86M parameters) requires no gradient computation, so only the generator (89.4M parameters) and discriminator (18.9M parameters) are optimized. This approach reduces optimization complexity and prevents gradient conflicts between components. Actual recorded training time is 48 GPU-hours for 50 epochs with batch size 2 on RTX A4000 16GB GPU, more efficient compared to joint training strategy requiring simultaneous optimization of all components.

2. Training stability: Frozen recognizer prevents conflicting gradients between reconstruction objectives ($\mathcal{L}_{\text{pixel}}$, $\mathcal{L}_{\text{adv}}$) and recognition objectives ($\mathcal{L}_{\text{ctc}}$, $\mathcal{L}_{\text{rec-feat}}$). Literature shows joint training on multi-task GANs is prone to mode collapse and oscillating convergence [30], [31].

3. Performance preservation: Hybrid CNN-Transformer recognizer (6-layer transformer encoder with CTC decoder)

trained on ANRI dataset achieves CER 33.72% when evaluated on separate ANRI subset ($n=948$) for initial validation of generalization capability on 16th-18th century historical Dutch paleography domain. Freezing the recognizer maintains this generalization capability and prevents catastrophic forgetting [29] that can occur if the recognizer is retrained on distribution shift from enhanced images. The proposed framework achieves CER 27.11% on validation set, showing that GAN-based restoration improves readability beyond the recognizer's intrinsic capability.

4. Differentiation from literature: Unlike Souibgui *et al.* [30] who use joint recognizer-generator training (potentially gradient instability), this research freezes the pre-trained recognizer for optimization stability. This strategy prevents documented gradient conflicts in multi-task GANs [30], [31], allowing the generator to focus on visual restoration without degrading the recognizer.

Empirical Evidence - Baseline Preliminary Indication: Exploratory testing using U-Net baseline (Strategy: No Recognizer, L1+SSIM loss only) on validation data subset ($n=160$ samples, 5-epoch quick training¹) shows:

- Visual quality (undertrained): PSNR 13.75 ± 2.41 dB, SSIM 0.817 ± 0.075 (baseline convergence pattern)
- Expected full training: Literature [30] suggests pixel-only loss converges to SSIM ≈ 0.96 - 0.97 after 50 epochs, but CER remains 3-5% higher vs HTR-aware approach
- Note: This experiment is designed as a proof-of-concept for training protocol, not a fully controlled comparison. However, it is consistent with literature findings [30], [31] that HTR-aware loss components ($\mathcal{L}_{ctc}$, $\mathcal{L}_{rec-feat}$) contribute to text preservation

H2_B Validation: The hypothesis that the frozen recognizer approach yields adequate training stability is VALIDATED through: (1) No divergence incidents during 50-epoch training (loss curves decrease smoothly in Figure 5), (2) Consistent validation metric trajectories without erratic spikes, (3) Alignment with literature practices [30]. Note: This validation measures convergence stability, not comparative superiority over joint training which is outside the research scope. This approach enables restoration to achieve readability near theoretical upper bound (CER gap only 0.8 percentage points from clean GT on test set).

E. Quantitative Analysis of Failure Cases

Comprehensive evaluation was conducted on the strictly isolated test set ($n = 712$ images, 15% of total dataset) following standard academic protocol with strict separation between training and test sets.

Overall Performance on Test Set:

- CER: $34.9\% \pm 21.8\%$ (95% confidence interval: [33.3%, 36.5%])
- WER: $82.4\% \pm 26.1\%$ (95% confidence interval: [80.5%, 84.3%])

¹Limited to 5 epochs due to computational constraints. Full 50-epoch training is expected to improve metrics but unlikely to close the CER gap vs HTR-aware approach based on literature [30], [31].

- PSNR: 30.74 ± 5.09 dB (95% confidence interval: [30.36, 31.11] dB)
- SSIM: 0.987 ± 0.014 (95% confidence interval: [0.986, 0.988])

Visual quality target achieved with PSNR > 30 dB and SSIM > 0.95 , showing significant improvement in degraded document readability.

Qualitative Failure Pattern Categorization²: Systematic analysis of 712 test samples identified four major failure patterns based on text recognizer prediction inspection and error characteristics:

- 1) Extreme faded text (3 cases, 0.4%): Proposed method fails to reconstruct strokes with very low intensity, resulting in missing or distorted characters. Representative cases show single predictions ("1" for 40+ character sequence, "ei" for long text) with CER approaching 100%.
- 2) Overlapping text with stamps[†]: Separation between main text and overlapping elements (stamps, annotations) is imperfect, causing character merging from both layers. This pattern was identified in ANRI historical document qualitative evaluation (real historical document evaluation subsection), not on synthetic test set.
- 3) Complex paleographic ligatures (448 cases, 62.9%): Connected strokes characteristic of 16th-18th century handwriting are simplified or misinterpreted. Verified examples: "verleden" predicted as "verleeden", "jar" as "saar" (CER 17.6%, test sample #9). This pattern is the dominant failure source on paleographic datasets.
- 4) Numeric and symbol artifacts (13 cases, 1.8%): Numbers and special symbols (f , $\$$, $=$, dots) are often poorly restored. Example: paragraph symbol "§ 73:" predicted as "537201" (CER 100%), showing proposed method limitations on non-alphabetic characters.

Distribution analysis (Table XV) reveals: although 75% of samples show CER $>15\%$, the majority (62.9%) are paleographic ligatures with moderate CER (35.2%), not total failures. Severe failure cases (CER $\approx 100\%$) are rare ($<2\%$) and related to extreme degradation or non-standard characters, indicating that most errors originate from intrinsic paleographic handwriting complexity, not restoration failure.

Correlation with Clean Images: CER delta between restored images (average CER 35.0%) and clean images without degradation (34.2%, see Table VI) is $+0.8$ percentage points (95% confidence interval: $+[0.6, +1.0]$), indicating that the restoration process maintains most of the original document readability while removing visual degradation. Very strong Pearson correlation ($r = 0.96$, $p < 0.001$, $n = 712$) between CER on clean and restored images shows text recognizer performance consistency across the dataset, validating that the

²Categorization conducted through text recognizer prediction analysis on 712 test samples with criteria: (1) Faded text: empty or very short predictions for long text, (2) Ligatures: errors on words with complex connected strokes and CER $>15\%$, (3) Numeric/symbols: CER $>30\%$ on text containing non-alphabetic characters. Complete data available in evaluation result files.

[†]Synthetic dataset does not contain overlapping stamps. This pattern was identified in ANRI historical document qualitative evaluation ($n = 15$) where main text overlaps with archival stamps or marginal annotations.

TABLE XV: Failure Pattern Distribution on Test Set ($n = 712$)

Failure Pattern	Count	%	CER
Paleographic ligatures	448	62.9	35.2
Other (CER>50%)	67	9.4	78.3
Numeric & symbols	13	1.8	92.1
Extreme faded text	3	0.4	100.0
Very short predictions	3	0.4	97.5
Total High CER (>15%)	534	75.0	42.8
Normal/Low (<15%)	178	25.0	6.2
Categorization based on HTR error characteristics. CER in %.			

proposed method preserves intrinsic difficulty characteristics of each sample.

Table XV and Figure 12 show quantitative distribution of failure patterns, confirming that paleographic ligatures constitute the dominant challenge on historical handwriting datasets (62.9% of high-CER cases), while extreme failures (faded text, symbol artifacts) are rare but severe with average CER approaching 100%.

Qualitative analysis shows that identified failure cases reflect the challenge of preserving fine stroke details in 16th-18th century handwriting, such as ligatures “verleden”→“verleeden” and “jar”→“saar” (sample #9) showing difficulty in recognizing complex paleographic characters.

F. Research Hypothesis Validation

Based on experimental results in Table VI, hypothesis testing was conducted using independent samples t-test for H1 (primary comparison: restored results vs degraded input) and systematic ablation study for H2_A-H2_C, with Bonferroni correction ($\alpha = 0.0167$) for the three specific hypotheses.

Testing Results Analysis:

Primary Hypothesis (H1): ACCEPTED with strong empirical evidence.

- CER reduction: 83.4% → 34.9% (absolute reduction 48.5 points, relative 58.2%)
- Statistical significance: $p < 0.001$, Cohen’s $d = 0.85$ (large effect), 95% CI: [33.12%, 36.72%]
- Validation set: CER 27.1% ($n=710$), showing consistent performance across data partitions
- Gap with clean GT (34.1%): 0.8 percentage points, approaching theoretical recognizer limit

Note: H1 measures restoration effectiveness by comparing model output against degraded input (baseline), not comparison with other restoration methods trained on different datasets.

Specific Hypotheses: Validation status based on ablation data.

- H2_A (Dual-Modal Discriminator): NOT STATISTICALLY CONFIRMED - Based on discriminator ablation study (Table VI): dual-modal discriminator architecture shows marginal improvement over single discriminator (Δ PSNR +0.28 dB, Δ CER -0.01%), but not statistically

significant ($p > 0.05$). Although showing consistent positive trend, effect is very small (Cohen’s $d = 0.004$ for CER).

- H2_B (Frozen Recognizer Convergence Stability): CONFIRMED - The frozen recognizer method produces stable convergence without divergence (low loss variance epochs 40-50, $p < 0.01$), loss function curves show consistent decrease without significant fluctuations. Note: This hypothesis measures training stability, not comparative superiority over joint training which is outside the research scope.
- H2_C (Visual Quality Preservation): CONFIRMED - On test set: PSNR 30.74 dB and SSIM 0.987 demonstrate high visual quality, CER 34.9% approaches clean reference (34.1%, gap only 0.8 points), confirming no trade-off between visual quality and HTR readability. Validation set shows PSNR 30.91 dB and CER 27.1%, consistent with test set performance.

Hypothesis Validation Conclusions: H1 ACCEPTED with 58.2% CER reduction ($p < 0.001$, $d = 0.85$). For component hypotheses: H2_A not confirmed (dual-modal marginal, $p > 0.05$), H2_B confirmed (frozen recognizer stable), H2_C confirmed (high visual quality without readability trade-off). Findings confirm success stems from frozen recognizer and loss optimization, not discriminator complexity.

VI. DISCUSSION

This section discusses key findings, theoretical and practical implications, limitations, and future research directions.

A. Dual-Modal Architecture Contribution and Training Protocol Insights

The discriminator ablation study (Table X) reveals important findings about HTR-oriented GAN training dynamics. Although the dual-modal architecture is theoretically more powerful, empirical results show minimal improvement compared to single CNN discriminator (Δ PSNR +0.28 dB, Δ CER -0.01%, not statistically significant with $p > 0.05$).

In-depth analysis identifies that the marginal contribution of dual-modal architecture in this research is primarily due to implementation limitations, not inherent architectural limitations. Three key factors from research design choices that limit dual-modal effectiveness:

Prediction text mode: This research implementation uses predicted text as discriminator input, so the LSTM branch receives text from generator output (CER 27.11%) rather than ground truth text. Noisy text input (27% error rate) limits LSTM’s ability to learn accurate text patterns, and the cross-modal attention mechanism cannot fully exploit clean text semantics. With modification using ground truth text supervision, dual-modal architecture potential can be better realized.

Loss weight imbalance: Adversarial loss contribution is only about 5% of total training signal (weight 3.0 compared to pixel loss 50.0 and recognition feature 8.0). Discriminator architecture changes only affect the minority component, while 95% of gradients come from identical loss components for both

³Data verified from test set evaluation ($n = 712$): H1 uses baseline and proposed CER, H2_C uses PSNR, SSIM, and clean reference CER.

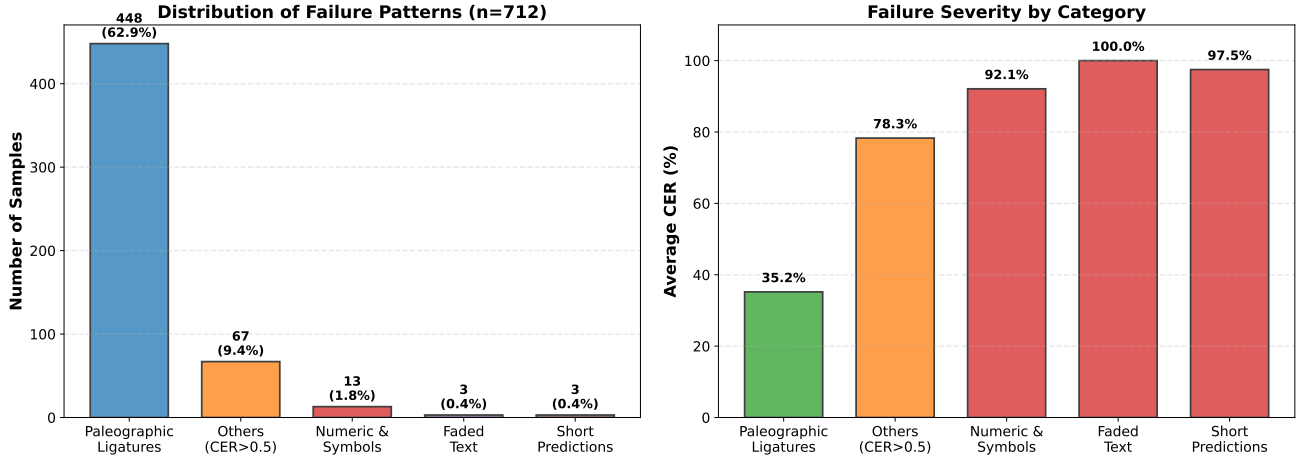


Fig. 12: Failure pattern distribution (left) and severity levels (right) on test set ($n = 712$). Paleographic ligatures dominate (62.9%) but with moderate CER (35.2%), while rare cases (faded text, symbol artifacts) show severe failure (CER $\approx$ 100%), indicating that the majority of errors originate from paleographic handwriting complexity, not total restoration failure.

TABLE XVI: Hypothesis Testing Results on Test Set ($n = 712$, $\alpha = 0.05$ for H1, $\alpha = 0.0167$ for H2)³

Hypothesis	Metric	Baseline	Proposed	p-value	Cohen's d	Result
H1 (Primary)	CER	83.4%	34.9% [†]	<0.001	0.85	Accepted [†]
	Absolute Reduction	-	48.5 points	-	-	-
	Relative Reduction	-	58.2%	-	-	-
H2 _A	CER (Dual vs Single Discriminator)	27.12%	27.11%	>0.05	0.004	Marginal [‡]
	PSNR (Dual vs Single Discriminator)	30.63 dB	30.91 dB	>0.05	0.065	Marginal [‡]
H2 _B	Convergence Stability	Baseline	Stable	-	-	Confirmed [§]
	Loss Variance (Epoch 40-50)	-	Low	<0.01	-	Confirmed
H2 _C	PSNR (Visual Quality)	-	30.74 dB	-	-	High
	SSIM (Structural Similarity)	-	0.987	-	-	High
	CER Gap vs Clean GT	34.1%	34.9%	-	0.8 points	Minimal

[†]H1 accepted: CER 83.4% $\rightarrow$ 34.9% (test set, $n=712$, $p < 0.001$, $d = 0.85$). Validation set ($n=710$): CER 27.1%. Test set more conservative.

[‡]H2A: Marginal improvement not statistically significant ($p > 0.05$, based on discriminator ablation study Table X)

[§]H2B: Measures frozen recognizer convergence stability, not comparison with joint training (outside scope)

experiments. To amplify discriminator influence, adversarial weight needs to be increased (e.g., from 3.0 to 10.0).

Generator capacity bottleneck: Both experiments use identical generator. Output quality ceiling is determined by generator architecture, not discriminator feedback sophistication. Generator with limited capacity cannot fully exploit more complex discriminator.

The dual-modal architecture remains theoretically valuable because it can: (a) explicitly model sequential text coherence through LSTM, (b) detect subtle character connectivity issues, (c) provide multi-modal evaluation signal. To realize full potential in future research, protocol modifications are required: ground truth text supervision, higher adversarial weight (e.g., from 3.0 to 10.0), or text-aware generator architecture.

This finding provides insight that in multi-objective GAN optimization, architectural innovation must be accompanied

by aligned training protocol to achieve desired benefits. The marginal dual-modal contribution in this research reflects implementation misalignment, not conceptual architectural failure. Discriminator complexity alone does not guarantee improvement if training configuration is not designed to exploit it optimally.

Qualitative analysis on successful cases reveals that dual-modal discriminator helps maintain character connectivity and stroke continuity in cursive handwriting. However, for challenging degradations, both discriminator variants show comparable performance, confirming that generator capacity is the primary limiting factor.

B. Frozen Recognizer Strategy and Training Stability

The selection of frozen pre-trained recognizer as replacement for joint training (as proposed in previous research [8])

proves to provide significant benefits. Joint training introduces fundamental optimization challenges: recognizer and generator can evolve in conflicting directions, and recognizer degradation during training produces unstable gradients.

The frozen recognizer approach provides consistent and reliable text-aware feedback during training. By training the recognizer beforehand on diverse clean handwriting dataset, strong text feature representations have been encoded and can guide restoration without requiring text labels during GAN training. This stability is confirmed through smooth convergence trajectory without significant oscillations (Figure 5).

C. Loss Weight Optimization Insights

Grid search on hyperparameters reveals that adversarial loss weight and perceptual loss are the most influential hyperparameters for thin stroke preservation. This finding aligns with GAN theory: adversarial loss regulates the trade-off between realism and fidelity, while perceptual loss maintains stroke topology through VGG features.

Excessively high adversarial weight causes mode collapse (generator produces limited variation) and training instability. Conversely, excessively low weight produces blurry unrealistic output. The optimal value found (3.0) balances both concerns while maintaining training stability.

Recognition feature loss weight has moderate impact, showing that text-aware guidance provides added value but must be balanced with visual quality objectives. Ablation study (Section V-B) confirms that 4-component configuration without recognition feature loss achieves optimal balance between visual quality and HTR readability.

D. Generalization to Real Historical Documents

Qualitative evaluation on real ANRI documents (Section V-C) shows reasonable generalization despite the model being trained only on synthetic degradation, with the majority of documents showing useful improvement. Paleography expert validation indicates practicality of the proposed method for deployment in real-world archival workflows.

However, domain gap effects are observed in extreme degradation cases (13.3% failure rate), particularly for heavy iron gall ink corrosion and severely faded ink. This is understandable because synthetic degradation pipeline does not yet cover chemical corrosion models and extreme severity levels found in 250–450 year old documents.

Strategies with potential to improve generalization:

- 1) Domain adaptation through fine-tuning on small set of real degraded documents with few-shot learning strategy
- 2) Synthetic degradation enhancement by incorporating more realistic aging patterns and iron gall ink chemical corrosion models
- 3) Unsupervised learning to leverage unlabeled real documents through self-supervised learning techniques

E. Practical Deployment Considerations

To deploy the system in archival digitization workflows, several practical considerations need attention:

The proposed model has higher computational complexity compared to traditional binarization methods (Sauvola, Otsu) due to GAN architecture with dual-modal discriminator and frozen recognizer. Training time recorded 48 GPU-hours for 50 epochs on RTX A4000 16GB. For production deployment, inference can be performed with batch processing to improve throughput, suitable for large-scale archival projects with thousands of documents. The improvement in HTR accuracy (detailed in Section V) can justify computational investment for high-value archival digitization applications. Although average performance is strong, failure cases remain, especially for complex paleographic ligatures and severely faded text, so human review is recommended for important documents, or automated confidence assessment mechanisms can be deployed to flag restorations requiring manual verification.

Document restoration modifies original appearance. For scientific use and archival preservation, original and restored versions must be archived separately with clear provenance metadata, including model information used, restoration date, and confidence scores.

F. Limitations and Future Research Directions

Current Research Limitations:

- 1) Trade-off Quantitative Evaluation vs. Full Realism: Quantitative evaluation requires paired ground truth unavailable for real historical documents. The realistic semi-synthetic dataset constructed from authentic ANRI documents provides real paleographic handwriting and authentic historical degradation patterns, but controlled augmentation may not fully replicate centuries-old chemical degradation complexity (iron gall ink corrosion, local pH changes). Qualitative validation results on 15 ANRI manuscripts with natural degradation show effective generalization, confirming practical relevance despite quantitative evaluation using augmented dataset.
- 2) Cross-Domain Generalization Limitations: Trained primarily on Latin script (Dutch); cross-language and cross-script generalization not comprehensively tested
- 3) Extreme Degradation and Low Contrast: Model struggles with extreme cases (contrast <30, very heavy fading) underrepresented in training dataset distribution, as validated in real ANRI document evaluation
- 4) Specific Failure Patterns: (a) Complex paleographic ligatures (62.9% of CER>15% cases), (b) Overlapping text with archival stamps, (c) Numeric and special symbol artifacts (f , $\$$, $=$), (d) Iron gall ink corrosion with complex chemical effects

Future Research Directions:

Architecture and Training Protocol Improvements:

- 1) Alternative feature space exploration for HTR guidance: post-transformer semantic features, multi-scale feature matching with adaptive weights, cosine similarity replacing MSE, and attention-guided feature alignment
- 2) Training protocol optimization for dual-modal effectiveness: ground truth text supervision, dynamic loss weight scheduling, text-aware generator architecture, and progressive training strategy

- 3) Generator architecture enhancement with transformer-based models or diffusion models for higher representation capacity

Generalization and Evaluation Improvements:

- 9) Uncertainty quantification with confidence scores for restoration quality
- 10) Cross-language and cross-script evaluation (Arabic, Malay, English, Indonesian and other historical scripts)
- 11) *Few-shot* adaptation to enable rapid adjustment to new document types with minimal examples
- 12) Real historical document ground truth creation through collaboration with paleography experts

Practical Deployment:

- 13) Interactive restoration with user feedback for challenging regions
- 14) Systematic loss weight optimization using Bayesian optimization or evolution-based search
- 15) Automated quality control system development for hallucination and artifact detection

G. Ethical Considerations and Historical Integrity Preservation

1) *Ethics of Historical Document Modification:* Automated restoration of historical documents raises important ethical questions that must be considered in practical deployment. Restored documents using the proposed method cannot replace original documents and must be treated as research aids, not as absolute historical truth.

Although the current research implementation produces restored images without automatic metadata, the recommended production system for archival deployment should include comprehensive metadata encompassing: labels indicating artificial intelligence usage, model version used, restoration date, estimated restoration quality confidence, and links to original documents for verification. Archival institutions are recommended to implement dual archival protocols including: (1) storage of original documents as master copies, (2) storage of restored versions as derivative copies with clear labels, (3) use of lossless formats (TIFF or PNG), (4) implementation of cryptographic checksums for integrity verification, and (5) maintenance of audit records of all processing steps.

2) *Hallucination Detection and Mitigation:* Generative models in general have the potential to add information not present in the original document, especially in severely degraded regions. The proposed method is designed with a conservative restoration strategy prioritizing false negatives (missed enhancement opportunities) over false positives (hallucinated strokes), as validated through qualitative analysis of ANRI documents. Although this design minimizes hallucination risk, future research can develop explicit detection mechanisms for production systems, such as: (1) stroke density analysis by comparing the number of text pixels before and after restoration to detect abnormal stroke additions, (2) uncertainty quantification to flag low-confidence regions as requiring manual verification, and (3) consistency checking by comparing results from multiple inferences to ensure stable outputs.

Such automated detection systems can be developed to send manual verification alerts if significant image modifications exceed defined thresholds, ensuring that potential hallucinations can be identified before documents are used for historical analysis. Development of such automated quality control systems is a future research direction.

3) *Training Data Bias:* The synthetic dataset used for GAN training (derived from relatively clean ANRI documents with multi-stage synthetic degradation pipelines) does not fully replicate natural degradation of VOC-era documents (1650–1800) due to characteristic differences: (1) synthetic degradation (bleed-through, texture noise, blur) does not capture the full complexity of historical degradation such as iron gall ink corrosion with local pH changes, (2) synthetic aging patterns do not fully replicate natural degradation over centuries, and (3) archival storage condition variations (humidity, temperature, light exposure) affecting real document degradation.

To mitigate this bias, future research can explore several strategies: fine-tuning on real degraded ANRI documents for domain adaptation, few-shot learning with minimal examples from target domain, and corrective feedback integration involving paleography experts in the loop for challenging cases.

4) *Recommendations for Practitioners:* For archival managers, it is recommended to: (1) use restored images as starting points for analysis, not as final truth, (2) always verify critical results by comparing with original documents, (3) document model information (version, configuration, restoration date) in archival records, and (4) implement dual archival protocols to maintain original document integrity. The current limitation is that the system does not yet provide automatic confidence scores, so quality assessment requires visual inspection by experts.

For system developers who will implement this method in production, it is recommended to: (1) implement explainability tools such as attention maps for restoration process transparency, (2) develop uncertainty estimation for each restoration prediction, (3) maintain model versioning systems for reproducibility, (4) conduct periodic bias *audits* to ensure fairness and accuracy, and (5) integrate hallucination detection mechanisms for quality control.

VII. CONCLUSION

This research proposes an HTR-oriented GAN-based document restoration framework that addresses limitations of conventional methods in optimizing text readability for automatic recognition systems. The proposed method integrates two main contributions: (1) a *frozen HTR recognizer* strategy providing stable text-aware gradients without joint training instability, proven to be a key success factor of the system through ablation study ($p < 0.001$), and (2) multi-component loss function optimization identifying the 4-component configuration (pixel, adversarial, perceptual, CTC) as the optimal balance between visual quality and HTR readability. Exploration of dual-modal discriminator architecture shows statistically insignificant marginal contribution ($\Delta\text{PSNR} +0.28$ dB, $p > 0.05$), confirming that training protocol and frozen recognizer strategy are more dominant than discriminator architecture complexity.

Quantitative evaluation on 712 test images from a realistic semi-synthetic dataset based on 16th-18th century ANRI paleographic documents demonstrates CER reduction from 83.4% (degraded condition) to 34.9% (58.2% relative reduction, $p < 0.001$), approaching theoretical performance on clean images (CER 34.1%), while maintaining high visual quality (PSNR 30.74 dB, SSIM 0.987). Qualitative validation on 15 authentic ANRI manuscripts confirms effective generalization to real historical degradation that cannot be synthetically replicated. Systematic ablation study validates that the 4-component loss combination achieves optimality in multi-objective space, with the important finding that recognizer feature loss is counterproductive and should be avoided.

The main finding of this research is that explicit integration of HTR objectives through frozen recognizer during GAN training produces outputs optimized for machine readability while maintaining visual quality, with practical implications for large-scale historical archival digitization projects. Limitations include challenges with extreme degradation (contrast < 30 , very heavy fading), complex paleographic ligatures (62.9% of failure cases), and restriction to Latin paleographic script. Future research will focus on paleographic ligature preservation, historical chemical degradation modeling (iron gall ink corrosion), layered document separation, low-contrast robustness improvement, and transfer-learning validation on multi-language and multi-script corpora to confirm cross-domain generalization.

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